# MATHEMATICS FOR COMMUNICATION ENGINEERING

(EC241AT)

**Notes Prepared by**

Dr. Mahesh A
Associate Professor, Department of ECE
RV College of Engineering

## Reference Textbooks

1. *Digital Communication Systems*, Simon Haykin, Wiley, Standard Edition, 2013
2. *Analog and Digital Communications*, Hwei P. Hsu, 2nd Edition,2003.
3. Lecture Notes of Dr. S. Ravishankar, Professor, Department of ECE, RVCE.

# Unit-4
# Digital Communication Transmitter

## Considerations for Line Codes:

1. ***Timing:*** The waveform produced by a line code should contain enough timing information such that the receiver can synchronize with the transmitter and decode the received signal properly. The timing content should be relatively independent of source statistics, i.e., a long string of 1s or 0s should not result in loss of timing or jitter at the receiver.

2. ***DC content:*** Since the repeaters used in telephony are ac coupled, it is desirable to have zero dc in the waveform produced by a given line code. If a signal with significant dc content is used in ac coupled lines, it will cause dc wander in the received waveform. That is, the received signal baseline will vary with time. Telephone lines do not pass dc due to ac coupling with transformers and capacitors to eliminate dc ground loops. Because of this, the telephone channel causes a droop in constant signals. This causes dc wander. It can be eliminated by dc restoration circuits, feedback systems, or with specially designed line codes.

3. ***Power spectrum:*** The power spectrum and bandwidth of the transmitted signal should be matched to the frequency response of the channel to avoid significant distortion. Also, the power spectrum should be such that most of the energy is contained in as small bandwidth as possible. The smaller is the bandwidth, the higher is the transmission efficiency.

4. ***Performance monitoring:*** It is very desirable to detect errors caused by a noisy transmission channel. The error detection capability in turn allows performance monitoring while the channel is in use (i.e., without elaborate testing procedures that require suspending use of the channel).

5. ***Probability of error:*** The average error probability should be as small as possible for a given transmitter power. This reflects the reliability of the line code.

6. ***Transparency:*** A line code should allow all the possible patterns of 1s and 0s. If a certain pattern is undesirable due to other considerations, it should be mapped to a unique alternative pattern.

## PSD of Line Codes

### Wiener–Khinchine Theorem

A line code is mathematically represented as

$$s(t) = \sum_{n=-N}^{N} a_n\, g(t - nT_b), \qquad N \to \infty \tag{1}$$

where $a_n$ is statistical and $g(t)$ is the lowpass pulse shaping function.

$$\mathcal{F}\{s(t)\} = S(f) = \sum_{n=-N}^{N} a_n\, G(f)\, e^{-j\omega n T_b} \tag{2}$$

$$S(f) = G(f) \sum_{n=-N}^{N} a_n e^{-j\omega n T_b} \tag{3}$$

PSD of a random signal $s(t)$ is

$$S_{xx}(f) = \frac{|S(f)|^2}{T} \tag{4}$$

For a random signal $s(t)$ with $T \to \infty$,

$$S_{xx}(f) = \lim_{T\to\infty} \left[ \frac{1}{T} |G(f)|^2 E \left| \sum_{n=-N}^{N} a_n e^{-j\omega n T_b} \right|^2 \right] \tag{5}$$

$$S_{xx}(f) = |G(f)|^2 \lim_{T\to\infty} \left[ \frac{1}{T} \sum_{n=-N}^{N} \sum_{m=-N}^{N} E[a_n a_m] e^{-j\omega(n-m)T_b} \right] \tag{6}$$

$$R(k) = E[a_n a_m] = E[a_n a_{n+k}] \tag{7}$$

$$R(k) = \sum_{i=1}^{L} (a_n a_{n+k})\, p_i \tag{8}$$

Let

$$T = (2N+1)T_b, \qquad N \to \infty, T \to \infty \tag{9}$$

$$S_{xx}(f) = |G(f)|^2 \lim_{N\to\infty} \left[ \frac{1}{(2N+1)T_b} \sum_{n=-N}^{N} \sum_{m=-N}^{N} E[a_n a_m] e^{-j(n-m)\omega T_b} \right] \tag{10}$$

Put

$$m = n + k, \qquad k = m - n \tag{11}$$

$$T = (2N+1)T_b \tag{12}$$

If $m = -N$ to $N$, then

$$k \to (-N - n) \text{ to } (N - n)$$

$$S_{xx}(f) = |G(f)|^2 \lim_{N\to\infty} \left[ \frac{1}{(2N+1)T_b} \sum_{n=-N}^{N} \sum_{k=-N-n}^{N-n} R(k) e^{jk\omega T_b} \right] \tag{13}$$

$$= \frac{|G(f)|^2}{T_b} \lim_{N\to\infty} \left[ \frac{1}{2N+1} \sum_{k=-N-n}^{N-n} R(k) e^{jk\omega T_b} \sum_{n=-N}^{N} (1) \right] \tag{14}$$

$$\boxed{S_{xx}(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k) e^{jk\omega T_b}} \tag{15}$$

where

$$R(k) = \sum_{i=1}^{L} (a_n a_{n+k})\, p_i \tag{16}$$

and $L$ is the number of levels.

# 1. Unipolar NRZ

$$s(t) = \begin{cases} A; \quad 0\ to\ T \quad \text{for } 1 \\ 0; \quad 0\ to\ T \quad \text{for } 0 \end{cases} \tag{17}$$

$$g(t) = \text{Rect}\left(\frac{t}{T_b}\right) \tag{18}$$

$$G(f) = T_b\,\text{sinc}(\pi f T_b) \tag{19}$$

$$R(k) = \sum_{i=1}^{L} (a_n a_{n+k}) p_i \tag{20}$$

For $k = 0$:

$$R(0) = \sum_{i=1}^{2} a_n a_n p_i = 0 \cdot 0 \cdot \frac{1}{2} + A \cdot A \cdot \frac{1}{2} = \frac{A^2}{2} \tag{21}$$

For $|k| > 0$:

$$R(1) = \sum_{i=1}^{L} (a_n a_{n+1}) p_i = \frac{A^2}{4} \tag{22}$$

Possible symbol pairs:

| $a_n$ | $a_{n+1}$ | $p_i$ |
|---|---|---|
| 0 | 0 | $\frac{1}{4}$ |
| 0 | 1 | $\frac{1}{4}$ |
| 1 | 0 | $\frac{1}{4}$ |
| 1 | 1 | $\frac{1}{4}$ |

$$S_{xx}(f) = \frac{|T_b \text{sinc}(\pi f T_b)|^2}{T_b} \left[ \frac{A^2}{2} + \sum_{k=-\infty, \neq 0}^{\infty} \frac{A^2}{4}\, e^{j2\pi k f T_b} \right] \tag{23}$$

$$= T_b\,\text{sinc}^2(\pi f T_b) \left[ \frac{A^2}{4} + \frac{A^2}{4} + \sum_{k=-\infty, \neq 0}^{\infty} \frac{A^2}{4}\, e^{j2\pi k f T_b} \right] \tag{24}$$

$$= \frac{A^2}{4} T_b\,\text{sinc}^2(\pi f T_b) \left[ 1 + \sum_{k=-\infty}^{\infty} e^{j2\pi k f T_b} \right] \tag{25}$$

Using the impulse train identity:

$$\sum_{k=-\infty}^{\infty} e^{j2\pi k f T_b} = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T_b}\right) \tag{26}$$

$$S_{xx}(f) = \frac{A^2 T_b}{4}\,\text{sinc}^2(\pi f T_b) \left[ 1 + \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T_b}\right) \right] \tag{27}$$

At $f = \dfrac{n}{T_b},\ n \neq 0$,

$$S_{xx}(f) = \frac{A^2 T_b}{4} \text{sinc}^2(\pi f T_b) \left[1 + \frac{\delta(f)}{T_b}\right] \tag{28}$$

$$= \frac{A^2 T_b}{4} \left[\frac{\sin(\pi f T_b)}{\pi f T_b}\right]^2 \left[1 + \frac{\delta(f)}{T_b}\right] \tag{29}$$

## Advantages

- Easy to generate, only one power supply is required.

## Limitations

1. Waste of power due to DC level at $f = 0$.
2. Clock extraction is not at peak.
3. Long strings of 1s and 0s cause loss of synchronization.
4. No error detection capability.

## Polar NRZ

$$s(t) = \begin{cases} A, & 0 \; to \; T_b \quad \rightarrow 1 \\ -A, & 0 \; to \; T_b \quad \rightarrow 0 \end{cases} \tag{30}$$

$$g(t) = \text{Rect}\left(\frac{t}{T_b}\right) \tag{31}$$

$$R(k) = \sum_{i=1}^{L} (a_n . a_{n+k}) p_i \tag{32}$$

For $k = 0$:

$$R(0) = A^2 \tag{33}$$

$$\begin{array}{ccc} a_n & a_n & p_i \\ A & A & \frac{1}{2} \\ -A & -A & \frac{1}{2} \end{array} \quad \Rightarrow R(0) = A^2$$

For $|k| = 1$:

$$R(1) = 0 \tag{34}$$

| $a_n$ | $a_{n+1}$ | $p_i$ |
|---|---|---|
| $A$ | $A$ | $\frac{1}{4}$ |
| $-A$ | $-A$ | $\frac{1}{4}$ |
| $-A$ | $A$ | $\frac{1}{4}$ |
| $A$ | $-A$ | $\frac{1}{4}$ |

Thus,

$$R(k) = \begin{cases} A^2, & k = 0 \\ 0, & k \neq 0 \end{cases} \tag{35}$$

$$S_{xx}(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k) e^{j\omega k T_b} \tag{36}$$

$$S_{xx}(f) = A^2 T_b \text{sinc}^2(\pi f T_b) \tag{37}$$

### Advantages

- Reduced DC content and better error performance compared to unipolar.

### Limitations

1. No error detection capability.
2. Two power supplies are required.
3. Long strings of 0s and 1s result in loss of synchronization.

## 3. Manchester Signaling

$$S_1(t) = \begin{cases} A, & 0 \; to \; T_b/2 \\ -A, & T_b/2 \; to \; T_b \end{cases} \quad \text{Symbol 1} \tag{38}$$

$$S_2(t) = \begin{cases} -A, & 0 \; to \; T_b/2 \\ A, & T_b/2 \; to \; T_b \end{cases} \quad \text{Symbol 0} \tag{39}$$

$$R(k) = \sum_{i=1}^{L} (a_n.a_{n+k}) p_i \tag{40}$$

For $k = 0$:

$$R(0) = \sum_{i=1}^{2} (a_n.a_n) p_i \tag{41}$$

$$= \frac{1}{2}[A.A\frac{1}{2} + (-A).(-A)\frac{1}{2} + (-A).(-A)\frac{1}{2} + A.A\frac{1}{2}] \tag{42}$$

$$= \frac{1}{2}[\frac{A^2}{2} + \frac{A^2}{2} + \frac{A^2}{2} + \frac{A^2}{2}] \tag{43}$$

$$R(0) = A^2 \tag{44}$$

For $k \neq 0$:

$$R(k) = 0 \tag{45}$$

$$g(t) = \text{Rect}\left(\frac{t + T_b/4}{T_b/2}\right) - \text{Rect}\left(\frac{t - T_b/4}{T_b/2}\right) \tag{46}$$

$$G(f) = \frac{T_b}{2}\text{sinc}\left(\frac{\pi f T_b}{2}\right) e^{j\omega T_b/4} - \frac{T_b}{2}\text{sinc}\left(\frac{\pi f T_b}{2}\right) e^{-j\omega T_b/4} \tag{47}$$

$$G(f) = jT_b\,\text{sinc}\left(\frac{\pi f T_b}{2}\right) \sin\left(\frac{2\pi f T_b}{4}\right) \tag{48}$$

$$S_{xx}(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k) e^{j\omega k T_b} \tag{49}$$

$$S_{xx}(f) = A^2 T_b \text{sinc}^2\left(\frac{\pi f T_b}{2}\right) \sin^2\left(\frac{\pi f T_b}{2}\right) \tag{50}$$

## Advantages

- Zero DC component.
- No loss of synchronization.

## Limitations

- No error detection capability.
- Large bandwidth required due to transitions.

# 4. AMI (Alternate Mark Inversion)

$+A$, $-A$ and 0 are used to represent AMI. Also called bipolar. Here we use RZ format.

$$g(t) = \text{Rect}\left(\frac{t}{T_b/2}\right) \tag{51}$$

$$= \frac{T_b}{2} Sinc(\frac{\pi f T_b}{2}) \tag{52}$$

$$R(k) = \sum_{i=1}^{L} (a_n.a_{n+k}) p_i \tag{53}$$

For $k = 0$:

$$R(0) = \frac{1}{2}(1.1) + \frac{1}{2}(0.0) \tag{54}$$

$$R(0) = \frac{1}{2}[\frac{1}{2}A.A + \frac{1}{2}(-A).(-A)] \tag{55}$$

$$R(0) = \frac{A^2}{2} \tag{56}$$

For $|k| = 1$:

$$\begin{array}{cccc} a_n & a_{n+1} & p_i & \\ 0 & 0 & 1/4 & = 0 \\ 0 & 1 & 1/4 & = 0 \\ 1 & 0 & 1/4 & = 0 \\ A & -A & 1/4 * 1/2 & = -A^2/8 \\ -A & A & 1/4 * 1/2 & = -A^2/8 \\ & & & = -A^2/4 \end{array}$$

$$R(1) = -\frac{A^2}{4} \tag{57}$$

For $|k| > 1$:

$$R(k) = 0 \tag{58}$$

$$S_{xx}(f) = \frac{\left|\frac{T_b}{2}\text{sinc}\left(\frac{\pi f T_b}{2}\right)\right|^2}{T_b} \left[\frac{A^2}{2} - \frac{A^2}{4}e^{j\omega T_b} - \frac{A^2}{4}e^{-j\omega T_b}\right] \tag{59}$$

$$= \frac{A^2 T_b}{8}\text{sinc}^2\left(\frac{\pi f T_b}{2}\right)\left[1 - \frac{1}{2}(e^{j\omega T_b} + e^{-j\omega T_b})\right] \tag{60}$$

$$= \frac{A^2 T_b}{8}\text{sinc}^2\left(\frac{\pi f T_b}{2}\right)[1 - \cos(\omega T_b)] \tag{61}$$

Using identity:

$$1 - \cos(\omega T_b) = 2\sin^2\left(\frac{\omega T_b}{2}\right) \tag{62}$$

$$S_{xx}(f) = \frac{A^2 T_b}{4}\text{sinc}^2\left(\frac{\pi f T_b}{2}\right)\sin^2(\pi f T_b) \tag{63}$$

## Advantages

- DC component is zero.
- Has single error detection capability.

## Limitations

1. Loss of synchronization when long strings of zeros are transmitted.
2. Two power supplies are required.
3. Needs 3 dB more signal power than polar signaling for same performance.

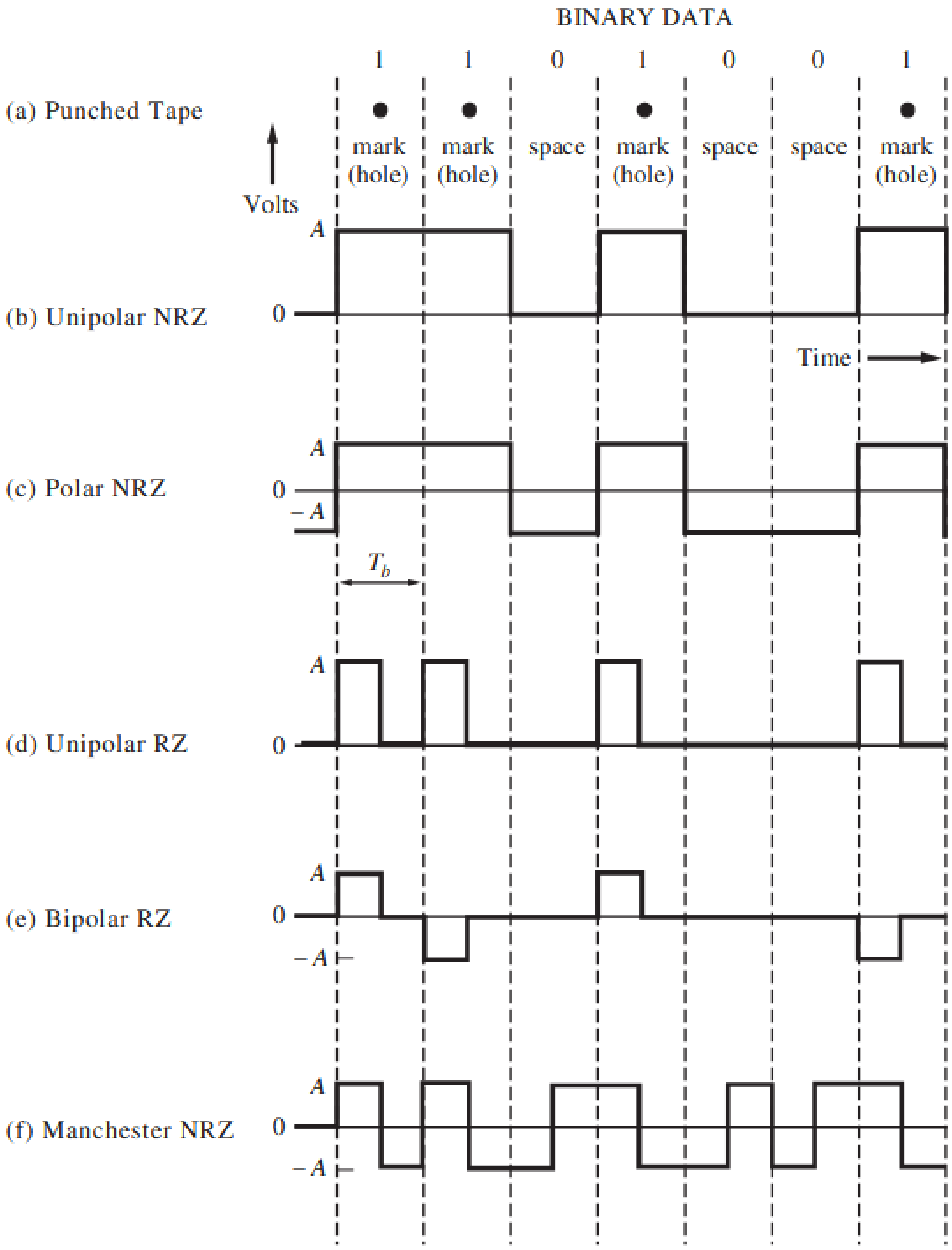



Figure 1: Binary Signalling formats

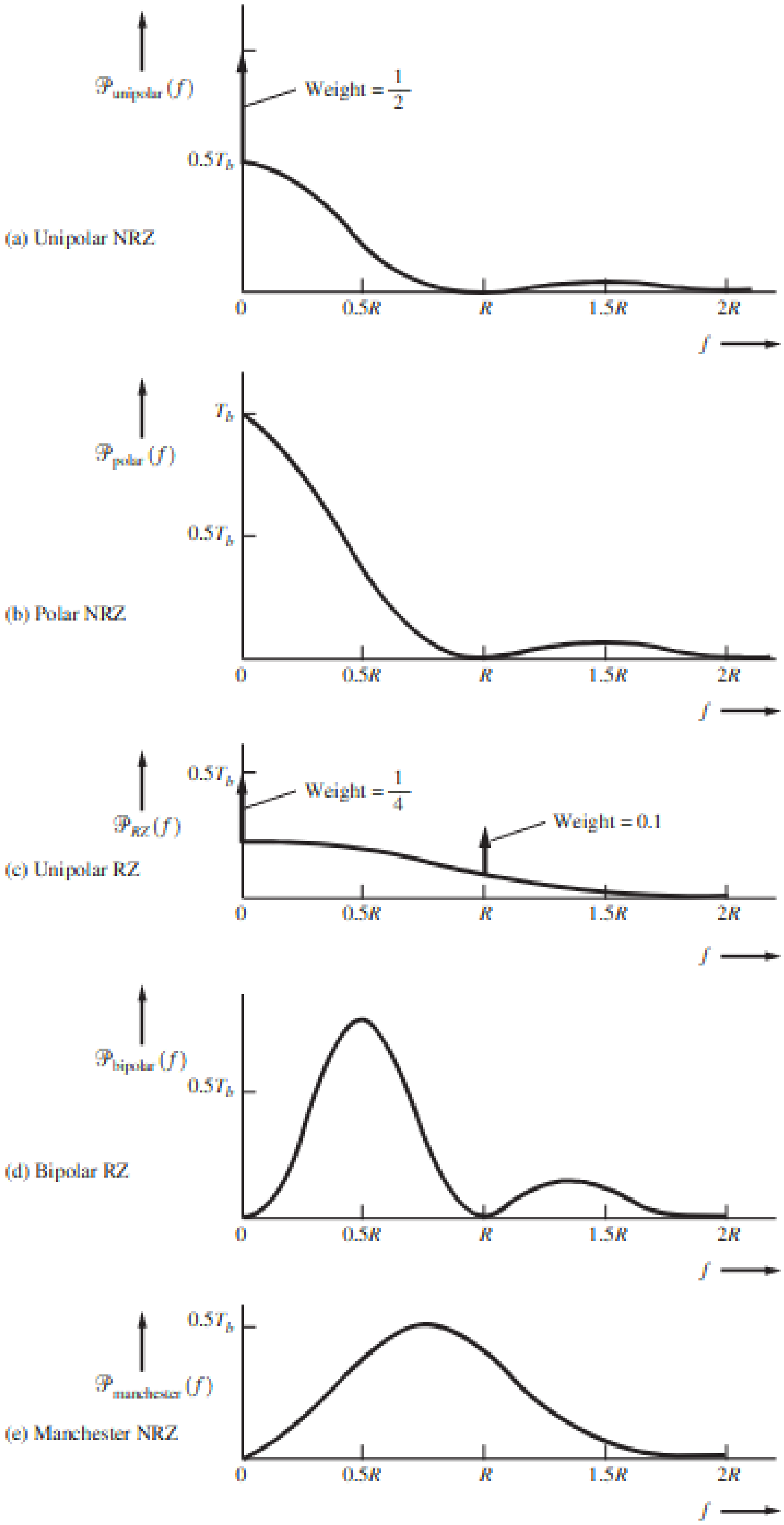
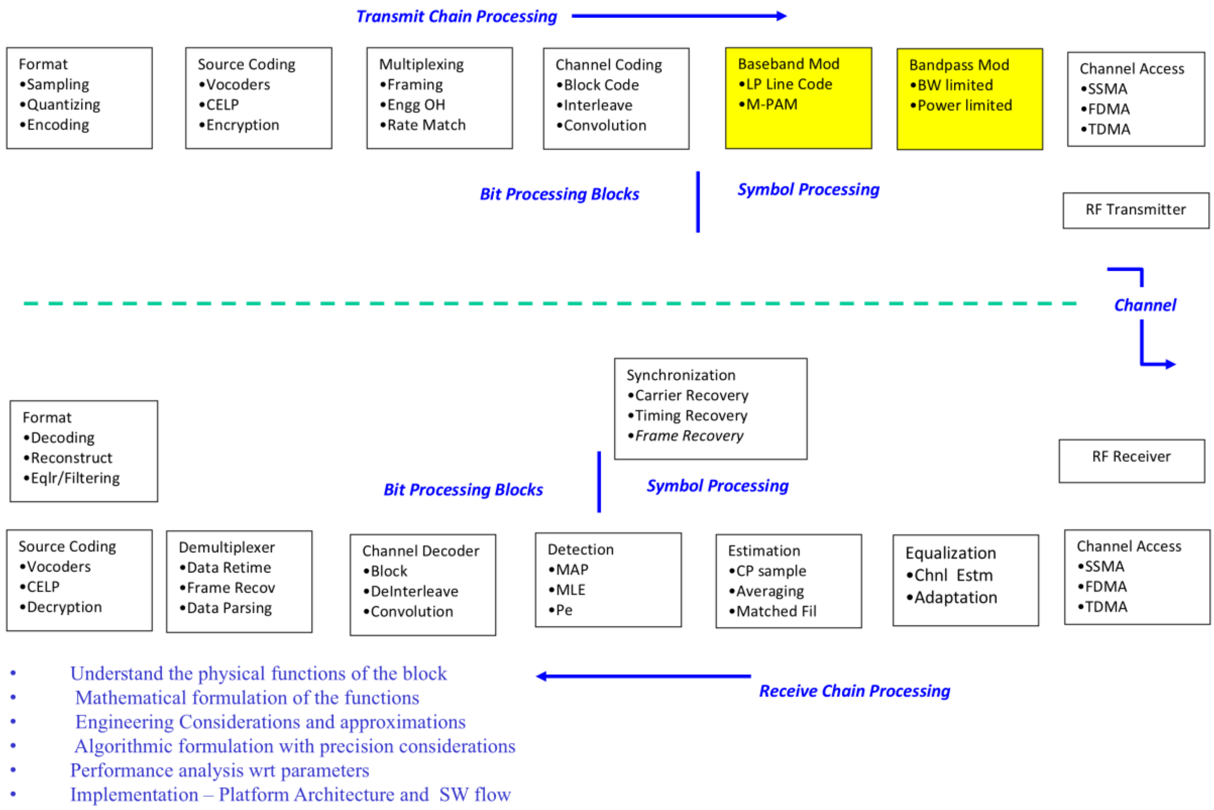


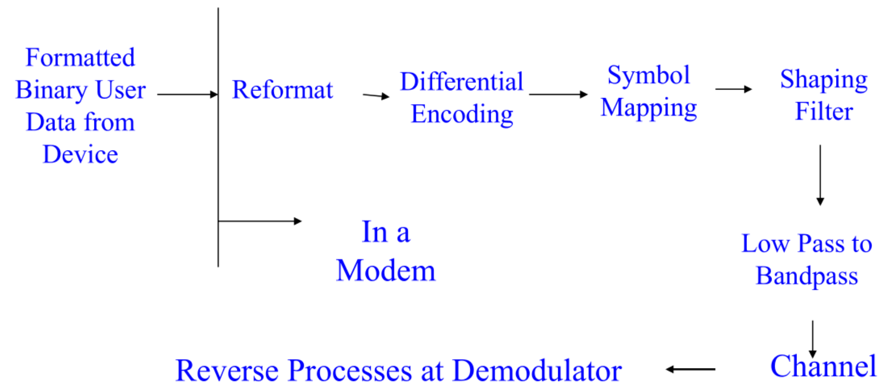

Figure 2: PSD for Line codes

# Bandpass Modulation




Modulation - Bit Stream to Bandpass Symbol Stream Conversion in 4 or 5 steps

- Utilizes mapping a group of bits to unique values of Amplitude, Phase, Frequency or combinations of the three parameters.
- The Signal is shaped so that spectra S(f) is channel friendly. Since multiple signals are employed we will need PSD of signal set.
- Translation to Bandpass frequencies



# Two classes of Modulation

1. **Modulation Scheme without Memory**

   Currently transmitted symbol has no signature of previous symbol.

   **Examples:** Polar, Unipolar, Manchester line codes, MPSK, MFSK, and MQAM.

2. **Modulation Scheme with Memory**

   Currently transmitted symbol has signature of previous symbols.

   **Examples:** AMI and its derivatives, Differential Modulation schemes, DPSK etc.

# A more important type of classification for Modulation that we intend to follow

## Bandlimited Modulation Schemes

- Increase rate by mapping more bits into symbols $\rightarrow$ increase number of symbols.
- For this we employ Amplitude and Phase only; NOT Frequency.
- Bandwidth is kept constant. Power is variable and increased for maintaining QOS (BER or SER).

  Bandwidth – Power Trade off applies. Coherently Demodulated.

**Examples:** PAM, PSK, QAM, MSK

## Power limited Modulation Schemes

- Increase rate by mapping more bits into more symbols i.e. increase number of symbols.
- For this we employ Frequency and Phase only; NOT Amplitude (linked to power).
- Transmitted power is constant. Bandwidth is variable and increased for maintaining QOS (BER or SER).

  Can be demodulated Coherently or Non-Coherently (frequency).

  Bandwidth – Power Trade off applies.

**Examples:** Orthogonal schemes – MFSK, MSK, ASK, DPSK, TSK (Time slot key)

## Spread Spectrum

- Bandwidth and Power can be varied independently.
- Spread spectrum margins are over the basic QOS, to be met by Power limited and Bandlimited schemes.

### Multicarrier Modulation

OFDM, DMT, Modal, FBMC, OTFS, NOMA

## Our study of Modulation will be in the order above and will encompass

- Mathematical Representation of Bandpass or low pass signal.
- Obtain the equivalent low pass representation.
- Obtain the basis functions for the low pass signal set.
- Express the low pass signal set in terms of low basis functions.
- Obtain the basis functions for the bandpass signals from low pass basis.
- Express the bandpass signals in terms of bandpass signal basis function.
- Draw the constellation diagram – Euclidean space representation of symbols using Bandpass basis functions / Real low pass basis for low pass signals as axes.
- Obtain distance between any two points in constellation and minimum distance $d_{\min}$.
- Obtain energy of symbols in the constellation and the average energy.
- Obtain the Transmit PSD.
- Transmitter Block Diagram.

## The described steps in the study of Modulation and Demodulation will be done as follows

1. Study of PAM, Bandpass PAM, PSK, QAM.
2. Study specific case of PAM – Line Codes (Unipolar, Polar, AMI, Manchester etc.) to obtain their PSD.
3. Study of Orthogonal Modulation schemes FSK, DPSK.
4. Unique Case MSK Modulation and Demodulation.
5. Coherent Demodulation for PSK and QAM encompassing:
    - Demodulation – Received symbol (corrupted + distorted) to Test Statistic.
    - Detection - Estimation – Test statistic to most possible symbol.
    - SER performance derivations.
    - Receiver Block Diagram.
    - Find the QOS - Probability of error.

6. Non-Coherent Demodulation for FSK and DPSK, SER performance and compare performance with Coherent demodulation.

# Constellation Diagram and Orthogonality

- Constellation diagram is a static Euclidean space representation (in time) of real (bandpass / lowpass) transmitted symbols with real orthonormal basis as axes.
- In Band limited schemes the SIGNALS $S_m(t)$ need not be orthogonal. But their basis functions are orthonormal.
- In Power limited schemes the SIGNALS $S_m(t)$ are usually maintained orthogonal and so are their basis function set.

## Choice of Symbols

Choice of symbols $\{s_m\}, \quad m = 1, 2, \ldots, M$

$$s_m(t) = \Re\left[s_{ml}(t)e^{j2\pi f_0 t}\right] \qquad \text{over } [0, T]$$

$$= \Re\left[A_m g(t)e^{j2\pi f_0 t}\right] \qquad \text{over } [0, T]$$

$$A_m \rightarrow \text{in general a complex number}$$

$$g(t) \rightarrow \text{Low pass shaped pulse that is real}$$

The values and expressions for $A_m$ are chosen as to realize varieties of symbols that are functions of amplitude, phase and frequency.

1. Set $f_0 = 0 \Rightarrow$ all symbols are low pass

$$f_0 = 0, \qquad A_m = \text{Real number}$$

$$\Rightarrow \text{PAM cases i.e. unipolar, polar etc.}$$

2. Set $f_0 \neq 0$

$$f_0 \neq 0, \qquad A_m = \text{Real number}$$

$$\Rightarrow \text{ASK}$$

Bandpass signal
(Carrier modulated PAM)

3. Set $f_0 \neq 0$, $A_m$ are complex

$$A_m = e^{j\frac{2\pi(m-1)}{M}}, \qquad m = 1, 2, \ldots, M$$

$$|A_m| = \text{constant}$$

$$\Rightarrow M\text{-PSK}$$

4. Set $f_0 \neq 0$, $A_m$ are complex

$$A_m = A_m^{(I)} + jA_m^{(Q)}$$

with

$$A_m^{(I)} \text{ and } A_m^{(Q)} \text{ real numbers}$$

$$m = 1, 2, \ldots, M$$

$$\Rightarrow M\text{-QAM}$$

5. Set $f_0 \neq 0$, $A_m$ are complex

$$A_m = e^{j2\pi m\Delta f\,t} = e^{j2\pi m\Delta Ft}$$

$$m = 1, 2, \ldots, M$$

$$\Rightarrow M\text{-FSK}$$

- Constellation diagram is a mapped Euclidean space representation (in time) of real (bandpass / lowpass) transmitted symbols with real orthonormal basis as axes.
- In Band limited schemes the SIGNALS $S_m(t)$ need not be orthogonal. But their basis functions are orthonormal.
- In Power limited schemes the SIGNALS $S_m(t)$ are orthogonal and so are their basis function set.

# Basis Functions and Representation of Symbols

## 1. M-PAM $(f_0 = 0$ case)

**Signal is lowpass and real**

$$s_m(t) = \Re\left[s_{ml}(t)e^{j2\pi f_0 t}\right]$$

Since

$$f_0 = 0$$

$$s_m(t) = \Re[s_{ml}(t)] = \Re[A_m p(t)]$$

Since $A_m$ are real,

$$s_m(t) = A_m p(t)$$

for any $m = 1, 2, \ldots, M$

$$g(t) = \frac{s_m(t)}{\sqrt{E_m}}$$

where

$$E_m = \int |s_m(t)|^2 dt = \int A_m^2 p^2(t)\, dt = A_m^2 E_p$$

Hence,

$$g(t) = \frac{s_m(t)}{A_m\sqrt{E_p}} = \frac{A_m p(t)}{A_m\sqrt{E_p}} = \frac{p(t)}{\sqrt{E_p}} = \phi_1(t)$$

over $[0, T]$

Thus only one lowpass basis exists since it is independent of $m$.

$$s_m(t) = A_m\sqrt{E_p}\,\phi_1(t)$$

for

$$m = 1, 2, \ldots, M$$

So we get to choose $A_m$ that are real.

$$2^k = M$$

$$A_m = (2m - 1 - M) \qquad m = 1, 2, \ldots, M$$

Amplitudes are

$$\pm 1, \pm 3, \pm 5, \ldots, \pm(M - 1)$$

## Choice of $A_m$

The choice of $A_m$ ensures

- No symbol at origin $\Rightarrow$ no confusion with noise.
- Being symmetric about origin in the constellation.
- Minimum average energy constellation.

For $M$ symbols

$$x_1, x_2, \ldots, x_M$$

Average symbol energy

$$E_{\text{avg}} = \frac{1}{M}\left(x_1^2 + x_2^2 + \cdots + x_M^2\right)$$

To change energy shift symbols by $a$

$$(x_1 - a), (x_2 - a), \ldots, (x_M - a)$$

Shifted average energy

$$E_{\text{shift}} = \frac{1}{M}\left[(x_1 - a)^2 + (x_2 - a)^2 + \cdots + (x_M - a)^2\right]$$

For minimum average energy

$$\frac{dE}{da} = 0$$

Hence,

$$2(x_1 - a) + 2(x_2 - a) + \cdots + 2(x_M - a) = 0$$

$$a = \frac{x_1 + x_2 + \cdots + x_M}{M}$$

Thus shifting has to be by mean of set

$$\{x_1, x_2, \ldots, x_M\}$$

New mean becomes zero.
Hence constellation is mean centered and symmetric about origin.

Therefore,

$$A_m = (2m - 1 - M) \qquad m = 1, 2, \ldots, M$$

Average energy becomes

$$E_{\text{avg}} = \frac{E_p}{M}\sum_{m=1}^{M} A_m^2$$

$$= \frac{E_p}{M}\left(1^2 + 3^2 + \cdots + (M-1)^2\right)$$

For symmetric constellation,

$$E_{\text{avg}} = \frac{(M^2-1)E_p}{3}$$

## Case: $f_0 \neq 0$

Signal is bandpass and real.

$$s_m(t) = \Re\left[s_{ml}(t)e^{j2\pi f_0 t}\right]$$

$$= \Re\left[A_m g(t)e^{j2\pi f_0 t}\right]$$

Lowpass signal

$$s_{ml}(t) = A_m g(t)$$

and

$$E_{sml} = \int_0^T A_m^2 g^2(t)\,dt = A_m^2 E_g$$

Hence

$$\phi_l(t) = \frac{s_{ml}(t)}{\sqrt{E_{sml}}} = \frac{A_m g(t)}{A_m\sqrt{E_g}}$$

$$\phi_l(t) = \frac{g(t)}{\sqrt{E_g}}$$

Thus only one lowpass basis exists.

Each lowpass basis function spans two bandpass basis functions

$$\phi_l^I(t) = \sqrt{2}\,\Re\left[\phi_l(t)e^{j2\pi f_0 t}\right]$$

$$\phi_l^Q(t) = -\sqrt{2}\,\Im\left[\phi_l(t)e^{j2\pi f_0 t}\right]$$

For this case,

$$\phi_l(t) = \frac{g(t)}{\sqrt{E_g}}$$

Hence,

$$\phi^I(t) = \sqrt{\frac{2}{E_g}}\,g(t)\cos(2\pi f_0 t)$$

$$\phi^Q(t) = -\sqrt{\frac{2}{E_g}}\, g(t)\sin(2\pi f_0 t)$$

# Bandpass Representation

Recall,

$$s_m(t) = \Re\left[s_{ml}(t)e^{j2\pi f_0 t}\right]$$

Also,

$$s_{ml}(t) = \sum_{n=1}^{N} s_{ml}^{(n)}\phi_n^{(l)}(t)$$

Substituting,

$$s_m(t) = \sum_{n=1}^{N}\left[\frac{s_{ml}^{(n)}}{\sqrt{2}}\phi_n^I(t) + \frac{s_{ml}^{(n)}}{\sqrt{2}}\phi_n^Q(t)\right]$$

Thus the coefficients of $s_{ml}(t)$ in $\sum s_{ml}^{(n)}\phi_n^{(l)}$ become as described above.

- The set of orthogonal basis functions constitute a Euclidean space for the class of symbols chosen (they are functions of $A, f, \phi$).
- The orthogonal basis set can be visualized as a set of rigid axes in $N$-dimensional space.
- If the symbols and their basis function are all real functions then the coefficients are also real.
- In this case the visualization can be as $N$-dimensional constellation as we see with examples below.
- This real symbols and real basis can be low pass or bandpass.

# 2. Carrier Modulated PAM (ASK)

**Signal is bandpass and real**

$$s_m(t) = A_m g(t)\cos(2\pi f_c t) \qquad [0, T]$$

$$= \Re\left[A_m g(t)e^{j2\pi f_c t}\right] = \Re\left[s_{ml}(t)e^{j2\pi f_c t}\right]$$

where

$$A_m \text{ is real and } g(t) \text{ is real}$$

and $A_m$ are chosen to be symmetric about origin.

Energy of lowpass signal:

$$E_{sml} = \int_0^T A_m^2 g^2(t)\, dt = A_m^2 E_g$$

Hence,

$$\phi_l(t) = \frac{s_{ml}(t)}{\sqrt{E_{sml}}} = \frac{A_m g(t)}{A_m \sqrt{E_g}} = \frac{g(t)}{\sqrt{E_g}}$$

Only one lowpass basis function exists.
Thus,

$$\phi_l(t) = \frac{g(t)}{\sqrt{E_g}}$$

and

$$s_{ml}(t) = A_m \sqrt{E_g}\, \phi_l(t)$$

Converting lowpass to bandpass basis:

$$\phi^I(t) = \sqrt{2}\Re\left[\phi_l(t)e^{j2\pi f_c t}\right]$$

$$= \sqrt{2}\Re\left[\frac{g(t)}{\sqrt{E_g}}e^{j2\pi f_c t}\right] = \sqrt{\frac{2}{E_g}}\, g(t)\cos(2\pi f_c t)$$

Similarly,

$$\phi^Q(t) = -\sqrt{2}\Im\left[\phi_l(t)e^{j2\pi f_c t}\right]$$

$$= -\sqrt{\frac{2}{E_g}}\, g(t)\sin(2\pi f_c t)$$

Now

$$s_{ml}(t) = s_{ml}^{(I)}(t) + j s_{ml}^{(Q)}(t)$$

For ASK,

$$s_{ml}^{(Q)}(t) = 0$$

Thus in lowpass equivalent domain,

$$s_{ml}(t) = (A_m \sqrt{E_g} + j0)\phi_l(t)$$

Hence in bandpass domain,

$$s_m(t) = \frac{s_{ml}^{(I)}}{\sqrt{2}}\phi^I(t) + \frac{s_{ml}^{(Q)}}{\sqrt{2}}\phi^Q(t)$$

Since

$$s_{ml}^{(Q)} = 0$$

$$s_m(t) = \frac{A_m\sqrt{E_g}}{\sqrt{2}}\phi^I(t)$$

Substituting for $\phi^I(t)$,

$$\begin{aligned} s_m(t) &= \frac{A_m\sqrt{E_g}}{\sqrt{2}}\sqrt{\frac{2}{E_g}}g(t)\cos(2\pi f_c t) \\ &= A_m g(t)\cos(2\pi f_c t) \end{aligned}$$

Constellation for Bandpass PAM lies only on the in-phase axis.

## Energy of Symbol

The energy of any symbol $m = 1, \ldots, M$ is $E_m$.

**First method:**
Directly from

$$s_m(t) = A_m g(t)\cos(2\pi f_c t)$$

$$E_m = \int_0^T s_m^2(t)\,dt$$

$$= \int_0^T A_m^2 g^2(t)\cos^2(2\pi f_c t)\,dt$$

Using

$$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$$

$$E_m = A_m^2\int_0^T g^2(t)\frac{1+\cos(4\pi f_c t)}{2}\,dt$$

$$= \frac{A_m^2}{2}\int_0^T g^2(t)\,dt + \frac{A_m^2}{2}\int_0^T g^2(t)\cos(4\pi f_c t)\,dt$$

For no spectral overlap,

$$\int_0^T g^2(t)\cos(4\pi f_c t)\,dt \approx 0$$

Hence,

$$E_m = \frac{A_m^2 E_g}{2}$$

**Second method:**

$$s_m(t) = A_m\sqrt{\frac{E_g}{2}}\phi^I(t)$$

Therefore,

$$E_m = \int_0^T s_m^2(t)\, dt$$
$$= A_m^2 \frac{E_g}{2} \int_0^T \left(\phi^I(t)\right)^2 dt$$

Since basis has unit energy,

$$\int_0^T \left(\phi^I(t)\right)^2 dt = 1$$

Hence,

$$E_m = \frac{A_m^2 E_g}{2}$$

## Average Symbol Energy

Average symbol energy is

$$E_{\text{avg}} = \sum_{m=1}^{M} A_m^2 \frac{E_g}{2} P(s_m)$$

For equiprobable symbols,

$$P(s_m) = \frac{1}{M}$$

Thus,

$$E_{\text{avg}} = \frac{E_g}{2M} \sum_{m=1}^{M} A_m^2$$

Using symmetric amplitudes,

$$A_m = \pm 1, \pm 3, \ldots, \pm(M-1)$$

$$E_{\text{avg}} = \frac{E_g}{2M} \left[1^2 + 3^2 + \cdots + (M-1)^2\right]$$

Using standard result,

$$1^2 + 3^2 + \cdots + (M-1)^2 = \frac{M(M^2-1)}{6}$$

Hence,

$$E_{\text{avg}} = \frac{(M^2-1)E_g}{6}$$

Energy per bit:

$$E_{b,\text{avg}} = \frac{E_{\text{avg}}}{k} = \frac{E_{\text{avg}}}{\log_2 M}$$

$$E_{b,\text{avg}} = \frac{(M^2-1)E_g}{6\log_2 M}$$

## Minimum Euclidean Distance

Euclidean distance between any pair of signal points:

$$d_{mn} = \sqrt{\|s_m - s_n\|^2}$$

For baseband PAM,

$$d_{mn} = \sqrt{(A_m - A_n)^2 E_p} = |A_m - A_n|\sqrt{E_p}$$

For bandpass PAM,

$$d_{mn} = \sqrt{(A_m - A_n)^2 \frac{E_g}{2}}$$

$$= |A_m - A_n|\sqrt{\frac{E_g}{2}}$$

For symmetric constellation,

$$A_m - A_n = 2$$

for adjacent symbols.
Hence,

$$d_{\min} = 2\sqrt{E_p} \qquad \text{(baseband PAM)}$$

and

$$d_{\min} = 2\sqrt{\frac{E_g}{2}} = \sqrt{2E_g} \qquad \text{(bandpass PAM)}$$

Expressing in terms of average energy:

$$E_p = \frac{3\log_2 M}{M^2-1}E_{b,\text{avg}}$$

and

$$E_g = \frac{6\log_2 M}{M^2-1}E_{b,\text{avg}}$$

Substituting,

$$d_{\min} = \sqrt{\frac{12\log_2 M}{M^2-1}E_{b,\text{avg}}}$$

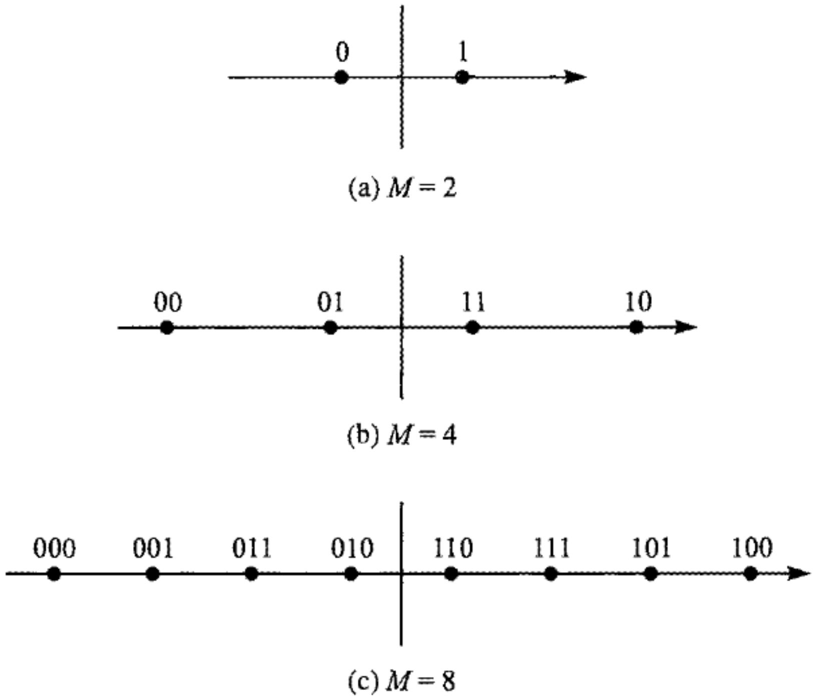



Figure 3: PAM Constellation Diagram

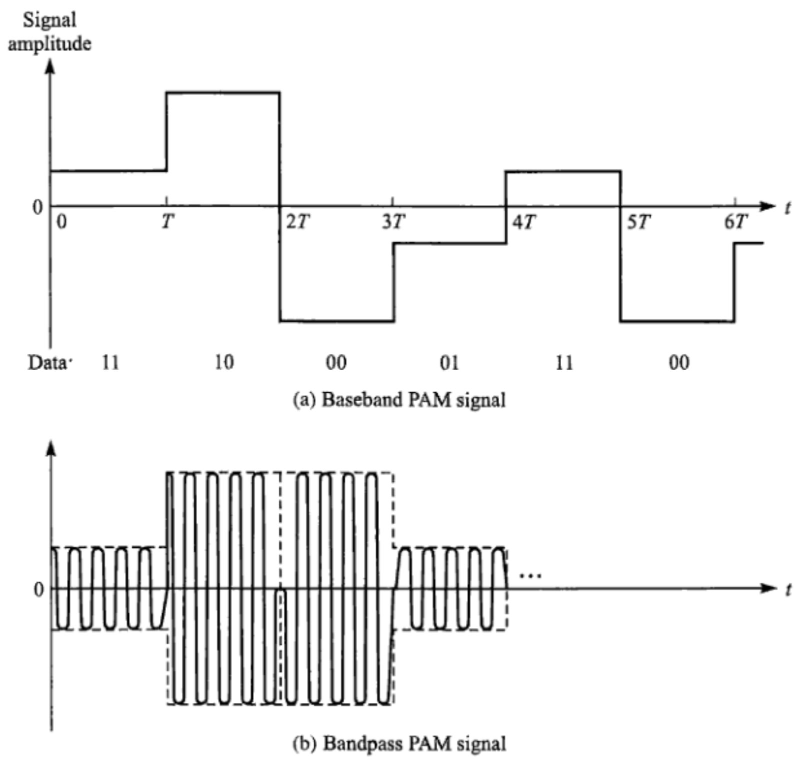



Figure 4: Example of (a). Baseband and (b). Carrier modulated signals

### Observations

- Linking $d_{\min}$ with average bit or symbol energy facilitates controlling transmitter power and thereby probability of SER or BER.

- SER and BER depend on $d_{\min}$ (difference between symbols) and not directly on individual symbol energies.
- Increase of $d_{\min}$ is achieved by increasing average energy which indirectly increases all individual symbol energies.
- Choice of symmetric constellations ensures that average energy is minimum for a required $d_{\min}$.

# 3. $M$-PSK Case

**Signal is bandpass and real**

$$f_0 \neq 0, \qquad A_m \text{ is complex}$$

$$s_m(t) = \Re\left[A_m g(t) e^{j2\pi f_0 t}\right] \qquad [0, T]$$

Choose

$$A_m = e^{j\frac{2\pi(m-1)}{M}} \qquad m = 1, 2, \ldots, M$$

Thus,

$$s_m(t) = \Re\left[e^{j\frac{2\pi(m-1)}{M}} g(t) e^{j2\pi f_0 t}\right]$$

Since

$$|A_m|^2 = A_m A_m^* = 1$$

all symbols have constant energy.

Lowpass signal:

$$s_{ml}(t) = A_m g(t)$$

and

$$E_{sml} = \int |A_m g(t)|^2 dt = |A_m|^2 E_g = E_g$$

Lowpass basis:

$$\phi_l(t) = \frac{s_{ml}(t)}{\sqrt{E_{sml}}} = \frac{A_m g(t)}{\sqrt{E_g}}$$

Since $A_m$ is complex,

$$\phi_l(t) = \frac{g(t)}{\sqrt{E_g}}$$

Hence only one lowpass basis exists.

Thus,

$$s_{ml}(t) = A_m \sqrt{E_g}\, \phi_l(t)$$

Bandpass basis functions:

$$\phi^I(t) = \sqrt{2}\Re\left[\phi_l(t)e^{j2\pi f_0 t}\right] = \sqrt{\frac{2}{E_g}} g(t)\cos(2\pi f_0 t)$$

$$\phi^Q(t) = -\sqrt{2}\Im\left[\phi_l(t)e^{j2\pi f_0 t}\right] = -\sqrt{\frac{2}{E_g}} g(t)\sin(2\pi f_0 t)$$

Now,

$$s_{ml}(t) = A_m \sqrt{E_g}\phi_l(t)$$

$$= e^{j\frac{2\pi(m-1)}{M}} \sqrt{E_g}\phi_l(t)$$

Hence,

$$s_{ml}^{(I)} = \sqrt{E_g}\cos\left(\frac{2\pi(m-1)}{M}\right)$$

$$s_{ml}^{(Q)} = \sqrt{E_g}\sin\left(\frac{2\pi(m-1)}{M}\right)$$

Therefore bandpass representation becomes

$$s_m(t) = \frac{s_{ml}^{(I)}}{\sqrt{2}}\phi^I(t) + \frac{s_{ml}^{(Q)}}{\sqrt{2}}\phi^Q(t)$$

$$= \sqrt{\frac{E_g}{2}}\cos\left(\frac{2\pi(m-1)}{M}\right)\phi^I(t) + \sqrt{\frac{E_g}{2}}\sin\left(\frac{2\pi(m-1)}{M}\right)\phi^Q(t)$$

Hence the constellation points lie on a circle.

## Euclidean Distance

Distance between two signal points:

$$d_{mn} = \sqrt{\|s_m - s_n\|^2}$$

Coordinates of symbol points:

$$s_m \to \left(\sqrt{\frac{E_g}{2}}\cos\frac{2\pi(m-1)}{M}, \sqrt{\frac{E_g}{2}}\sin\frac{2\pi(m-1)}{M}\right)$$

$$s_n \to \left(\sqrt{\frac{E_g}{2}}\cos\frac{2\pi(n-1)}{M}, \sqrt{\frac{E_g}{2}}\sin\frac{2\pi(n-1)}{M}\right)$$

Hence,

$$d_{mn} = \sqrt{\frac{E_g}{2}\left(\cos\theta_m - \cos\theta_n\right)^2 + \frac{E_g}{2}\left(\sin\theta_m - \sin\theta_n\right)^2}$$

where

$$\theta_m = \frac{2\pi(m-1)}{M}$$

Simplifying,

$$d_{mn} = \sqrt{E_g\left[1 - \cos(\theta_m - \theta_n)\right]}$$

Minimum distance occurs for

$$|m - n| = 1$$

Thus,

$$d_{\min} = \sqrt{E_g\left(1 - \cos\frac{2\pi}{M}\right)}$$

Using

$$1 - \cos 2\theta = 2\sin^2\theta$$

$$d_{\min} = \sqrt{2E_g \sin^2\frac{\pi}{M}}$$

$$d_{\min} = \sqrt{2E_g}\sin\frac{\pi}{M}$$

## Distance in terms of Average Energy

All constellation points have same energy.

Hence,

$$E_{\text{avg}} = E_m = \frac{E_g}{2}$$

Thus,

$$E_g = 2E_{\text{avg}}$$

Substituting,

$$d_{\min} = \sqrt{2(2E_{\text{avg}})}\sin\frac{\pi}{M}$$

$$d_{\min} = 2\sqrt{E_{\text{avg}}}\sin\frac{\pi}{M}$$

Since

$$E_{\text{avg}} = kE_b = (\log_2 M)E_b$$

$$d_{\min} = 2\sqrt{(\log_2 M)E_b}\sin\frac{\pi}{M}$$

For large $M$,

$$\sin\frac{\pi}{M} \approx \frac{\pi}{M}$$

Hence,

$$d_{\min} \approx 2\sqrt{(\log_2 M)E_b}\frac{\pi}{M}$$

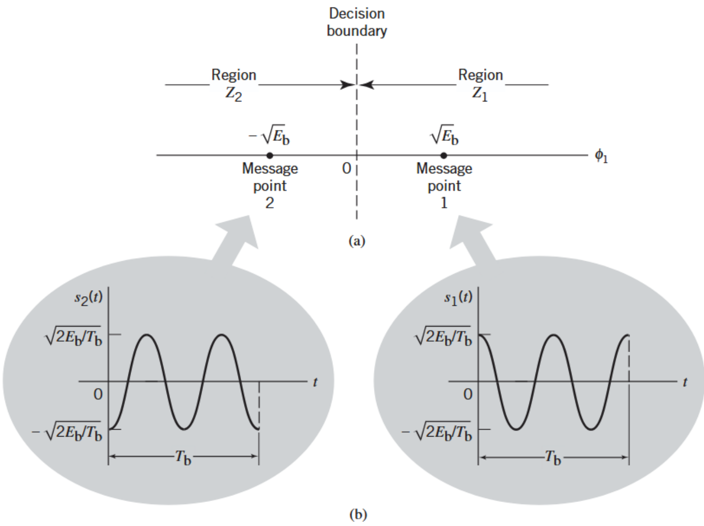



Figure 5: (a) Signal-space diagram for coherent binary PSK system. (b) The waveforms depicting the transmitted signals s1(t) and s2(t), assuming nc = 2

$$d_{12} = d_{min} = 2\sqrt{E_b} \tag{64}$$

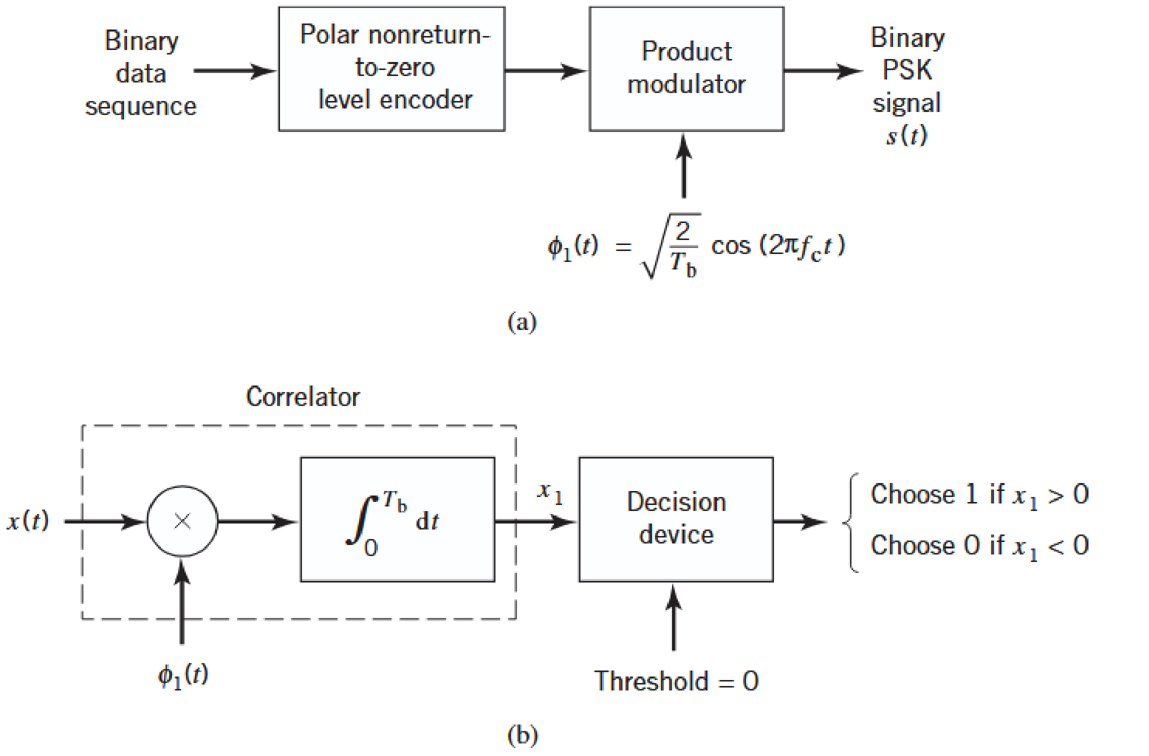



Figure 6: Block diagrams for (a) binary PSK transmitter and (b) coherent binary PSK receiver

## Power Spectral Density of BPSK (Baseband):

$$S_{xx}(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k)\, e^{jk\omega T_b} \tag{65}$$

Let $g(t)$ denote the pulse shaping function:

$$g(t) = \sqrt{\frac{2E_b}{T_b}}, \quad 0 \le t \le T_b \tag{66}$$

$$g(t) = \sqrt{\frac{2E_b}{T_b}}\, \mathrm{rect}\left(\frac{t}{T_b}\right) \tag{67}$$

$$G(f) = \sqrt{\frac{2E_b}{T_b}}\, T_b\, \mathrm{sinc}(fT_b) \tag{68}$$

$$S_{xx}(f) = \frac{2E_b/T_b}{T_b}\, T_b^2\, \mathrm{sinc}^2(fT_b) \tag{69}$$

$$S_{xx}(f) = 2E_b\, \mathrm{sinc}^2(fT_b) \tag{70}$$

$$S_{xx}(f) = 2E_s\, \mathrm{sinc}^2(fT_s) \tag{71}$$

Since for BPSK, $k = 1$:

$$E_s = E_b, \qquad T_s = T_b$$

In the above derivation, it was assumed that the incoming signal or sequence is random, with symbols 1 and 0 being equally likely and the symbols transmitted during different time slots being statistically independent.

## Notes

1. $S_{xx}(f)$ falls off as the inverse square of the frequency.
2. $S_{xx}(f)$ goes through zero at multiples of the bit rate.

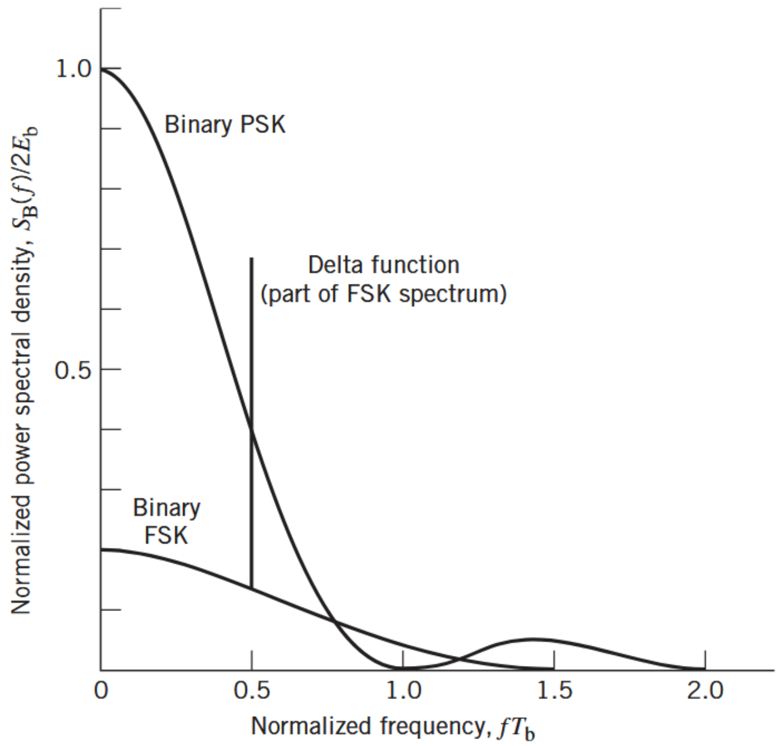



Figure 7: Power spectra of binary PSK and FSK signals

$$s_i(t) = \sqrt{\frac{2E}{T}} \left[ \cos(2\pi f_c t) \cos\left(\frac{(2i-1)\pi}{4}\right) - \sin(2\pi f_c t) \sin\left(\frac{(2i-1)\pi}{4}\right) \right] \tag{72}$$

PSD:

$$g(t) = \sqrt{\frac{E}{T}}, \quad 0 \leq t \leq T \tag{73}$$

Similar to BPSK, the PSD of QPSK is:

$$S_{xx}(f) = 2E_s \operatorname{sinc}^2(fT_s) \tag{74}$$

$$S_{xx}(f) = 2(2E_b) \operatorname{sinc}^2(2fT_b) \tag{75}$$

$$S_{xx}(f) = 4E_b \operatorname{sinc}^2(2fT_b) \tag{76}$$

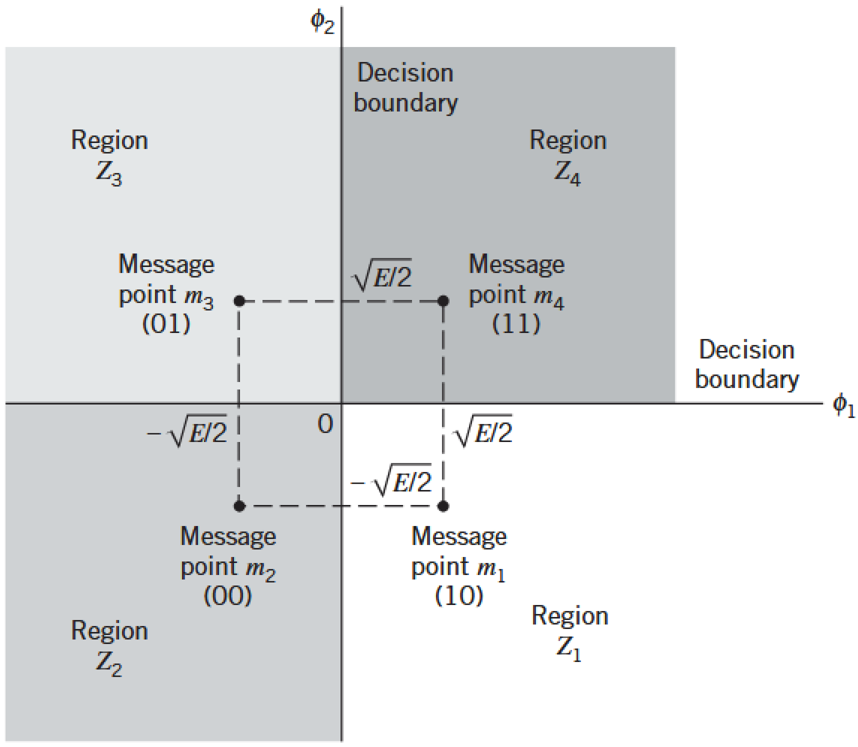



Figure 8: Signal-space diagram of QPSK system

$$d_{min} = d_{12} = d_{14} = \sqrt{2E_s} \tag{77}$$

| Gray-encoded input dibit | Phase of QPSK signal (radians) | Coordinates of message points | |
|---|---|---|---|
| | | $s_{i1}$ | $s_{i2}$ |
| 11 | $\pi/4$ | $+\sqrt{E/2}$ | $+\sqrt{E/2}$ |
| 01 | $3\pi/4$ | $-\sqrt{E/2}$ | $+\sqrt{E/2}$ |
| 00 | $5\pi/4$ | $-\sqrt{E/2}$ | $-\sqrt{E/2}$ |
| 10 | $7\pi/4$ | $+\sqrt{E/2}$ | $-\sqrt{E/2}$ |

Figure 9: Signal-space characterization of QPSK

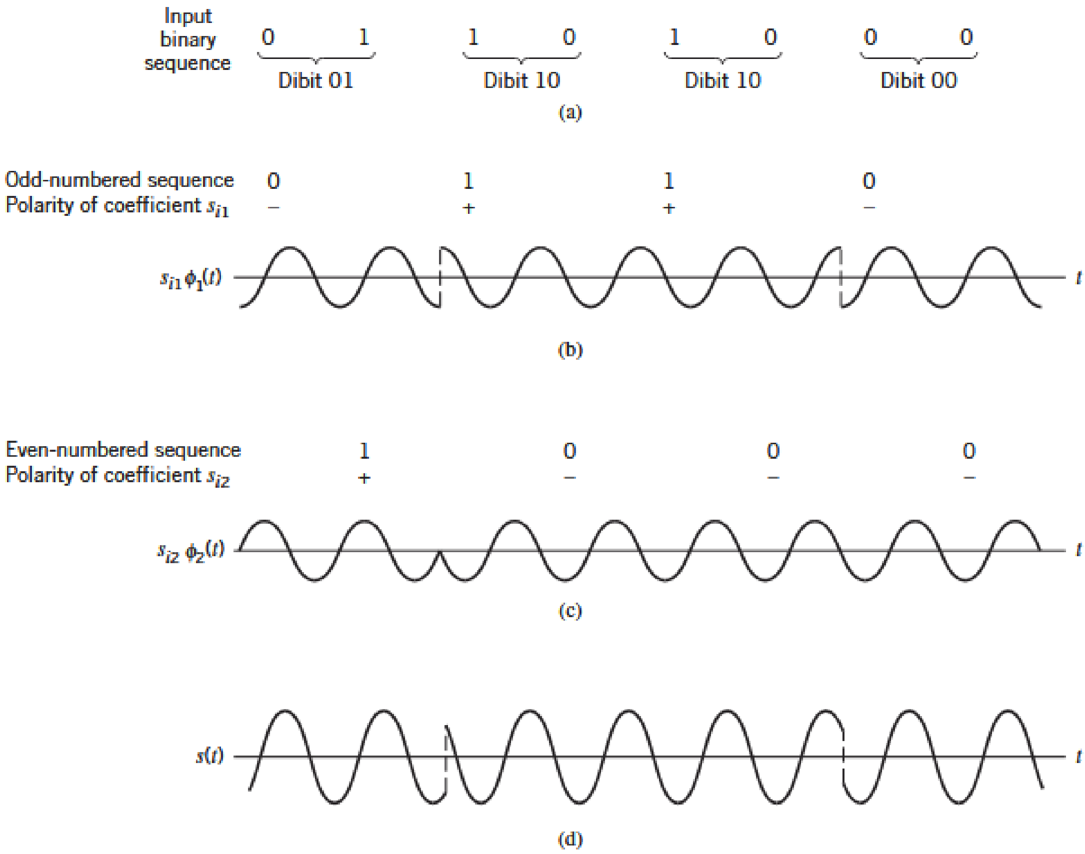



Figure 10: (a) Input binary sequence. (b) Odd-numbered dibits of input sequence and associated binary PSK signal. (c) Even-numbered dibits of input sequence and associated binary PSK signal. (d) QPSK waveform defined as

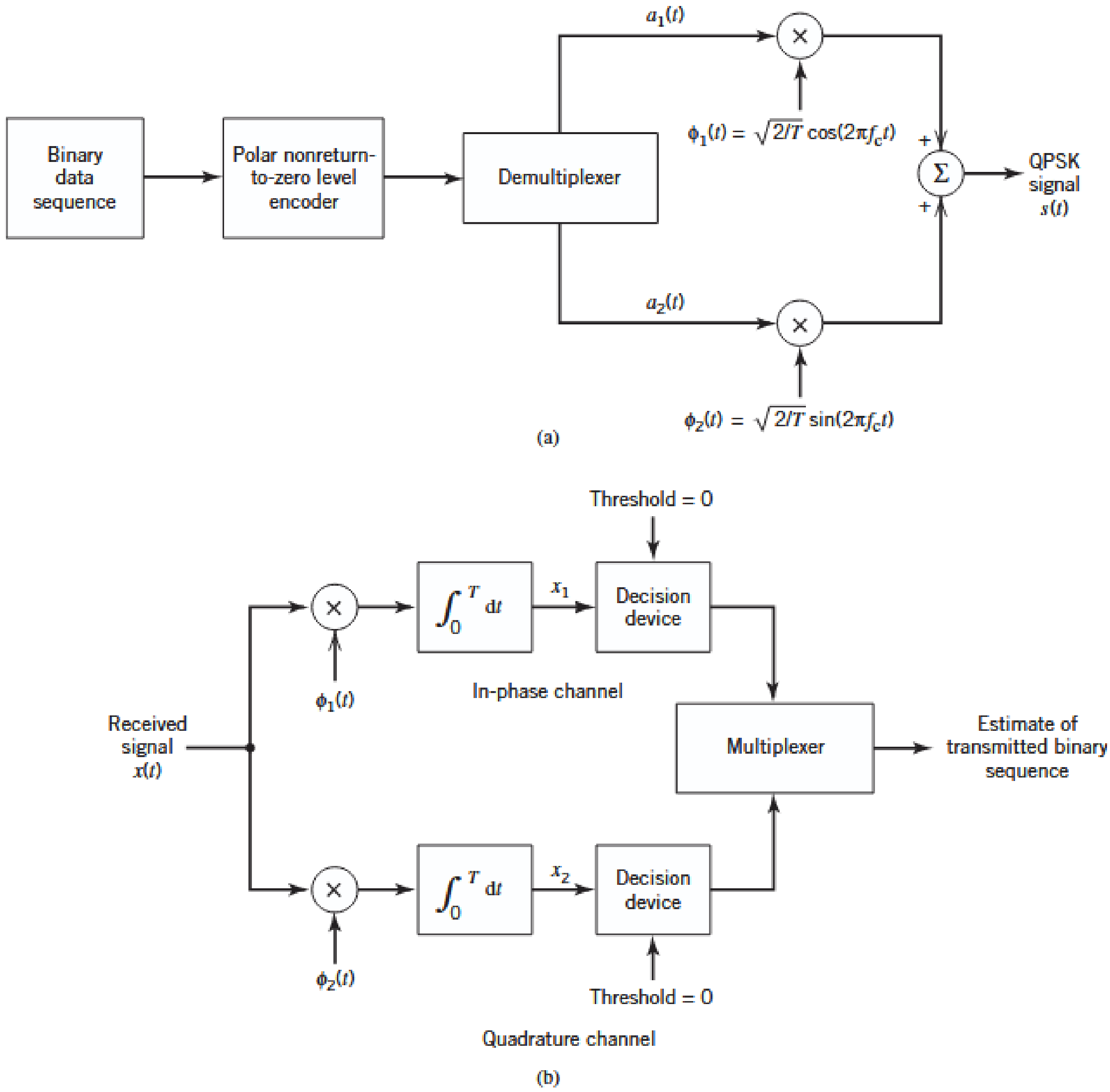



Figure 11: Block diagram of (a) QPSK transmitter and (b) coherent QPSK receiver

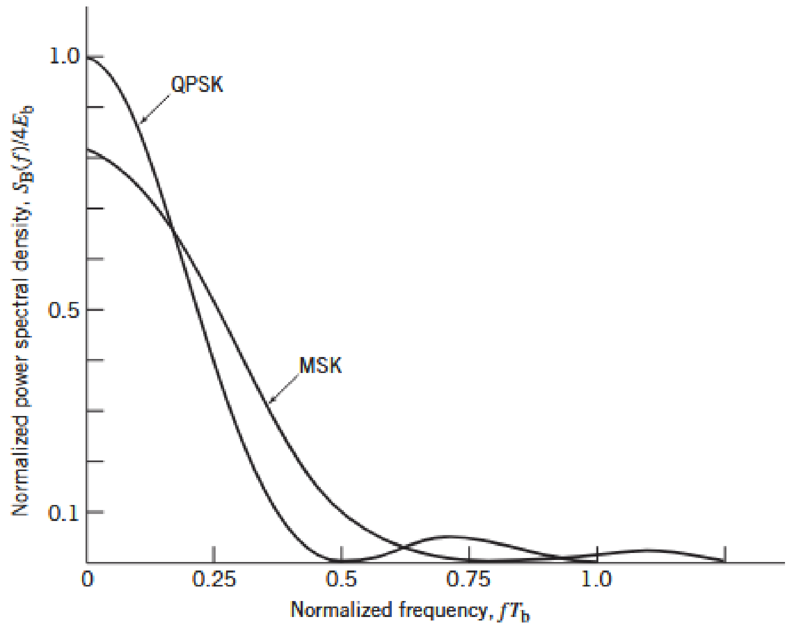



Figure 12: Power spectra of QPSK and MSK signals.

# 8-PSK:

$$k = 3, \quad M = 2^3 = 8$$

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos\left[2\pi f_c t + \frac{2\pi}{M}(i-1)\right], \quad i = 1, 2, \ldots, M \tag{78}$$

Two basis functions are sufficient to represent all 8 signals.

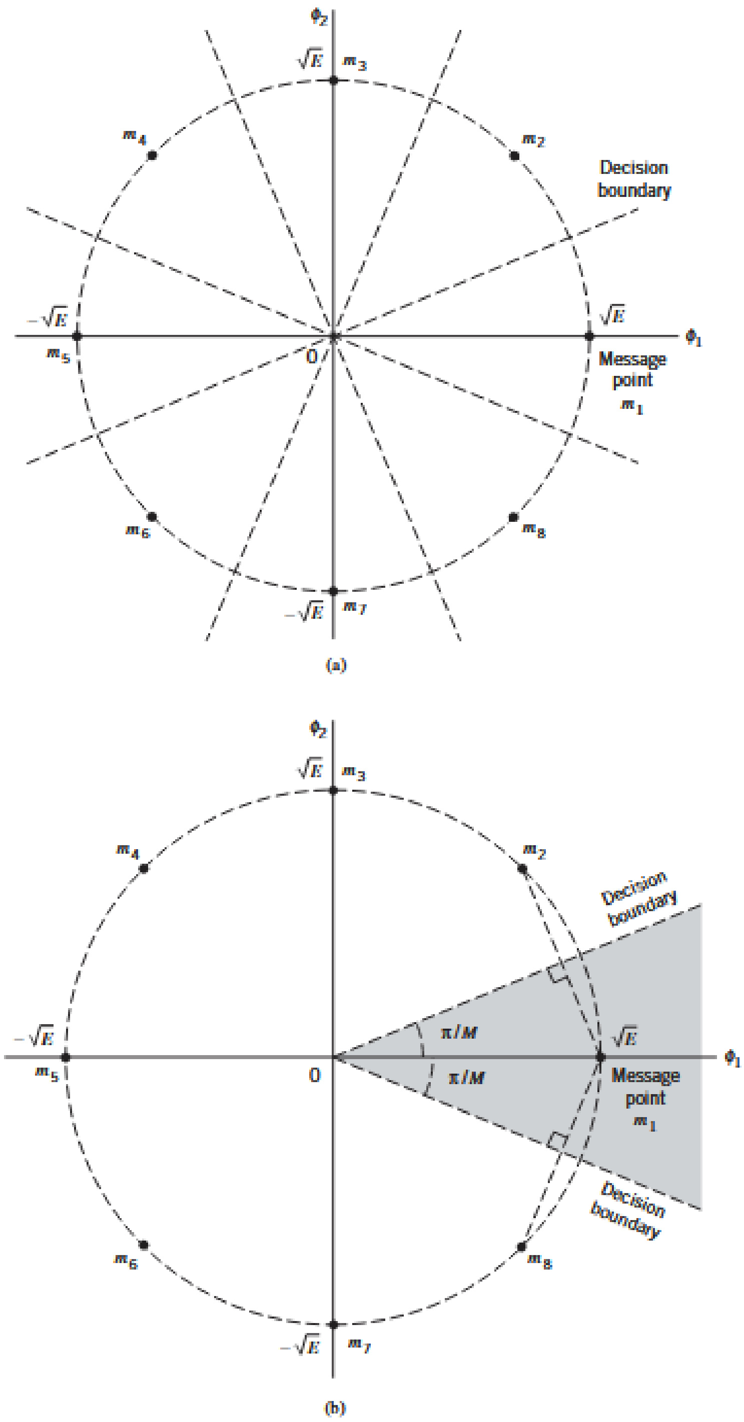



Figure 13: (a) Signal-space diagram for octaphase-shift keying (i.e., M = 8). The decision boundaries are shown as dashed lines. (b) Signal-space diagram illustrating the application of the union bound for octaphase-shift keying

## PSD Expressions for PSK Schemes

For $M$-PSK,

$$S_{xx}(f) = 2kE_b \operatorname{sinc}^2(fkT_b) \tag{79}$$

$$= 2E_b \log_2 M \operatorname{sinc}^2(fT_b \log_2 M) \tag{80}$$

For 8-PSK,

$$S_{xx}(f) = 6E_b \operatorname{sinc}^2(3fT_b) \tag{81}$$

For QPSK,

$$S_{xx}(f) = 4E_b \operatorname{sinc}^2(2fT_b) \tag{82}$$

For BPSK,

$$S_{xx}(f) = 2E_b \operatorname{sinc}^2(fT_b) \tag{83}$$

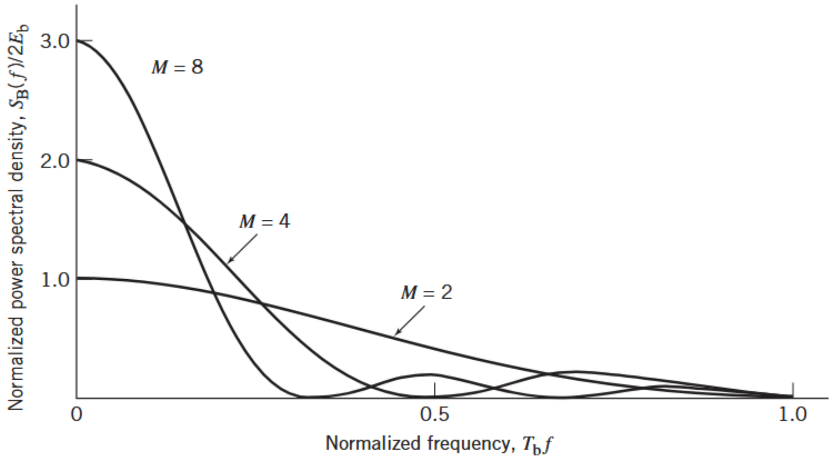



Figure 14: Power spectra of M-ary PSK signals for M = 2, 4, 8

# Offset QPSK (OQPSK)

Examining the QPSK waveform illustrated in Figure 6, we may make three observations:

1. The carrier phase changes by ±180° whenever both the in-phase and quadrature components of the QPSK signal change sign. An example of this situation is illustrated in Figure 4 when the input binary sequence switches from dibit 01 to dibit 10.

2. The carrier phase changes by ±90° whenever the in-phase or quadrature component changes sign. An example of this second situation is illustrated in Figure 4 when the input binary sequence switches from dibit 10 to dibit 00, during which the in-phase component changes sign, whereas the quadrature component is unchanged.

3. The carrier phase is unchanged when neither the in-phase component nor the quadrature component changes sign. This last situation is illustrated in Figure 4 when dibit 10 is transmitted in two successive symbol intervals.

Situation 1 and, to a much lesser extent, situation 2 can be of particular concern when the QPSK signal is filtered during the course of transmission, prior to detection. Specifically, the 180° and 90° shifts in carrier phase can result in changes in the carrier amplitude (i.e., envelope of the QPSK signal) during the course of transmission over the channel, thereby causing additional symbol errors at detection at the receiver.

To mitigate this shortcoming of QPSK, we need to reduce the extent of its amplitude fluctuations. To this end, we may use *offset QPSK*. In this variant of QPSK, the bit stream responsible for generating the quadrature component is delayed (i.e., offset) by half a symbol interval with respect to the bit stream responsible for generating the in-phase component. Specifically, the two basis functions of offset QPSK are defined by

$$\phi_1(t) = \sqrt{\frac{2}{T}}\cos(2\pi f_c t), \quad 0 \leq t \leq T \tag{84}$$

$$\phi_2(t) = \sqrt{\frac{2}{T}}\sin(2\pi f_c t), \quad \frac{T}{2} \leq t \leq \frac{3T}{2} \tag{85}$$

Since, in addition to ±90° phase transitions, ±180° phase transitions also occur in QPSK, we find that amplitude fluctuations in offset QPSK due to filtering have a smaller amplitude than in the case of QPSK.

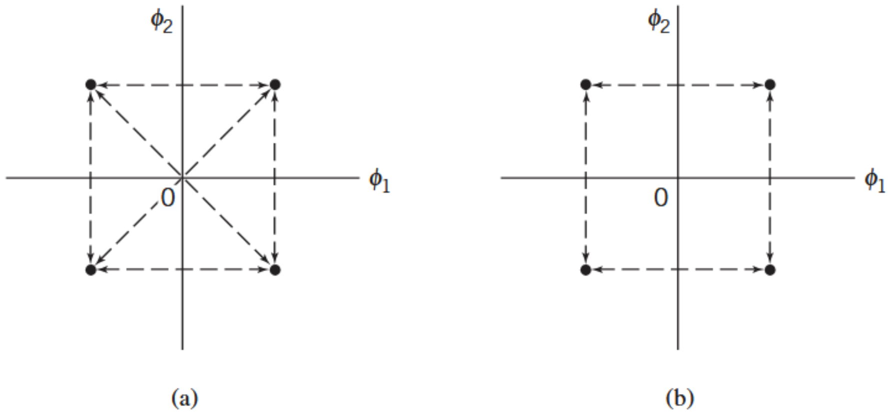



Figure 15: Possible paths for switching between the message points in (a) QPSK and (b) offset QPSK

# 4. QAM Case

$$f_0 \neq 0, \qquad A_m \text{ is complex}$$

**Signal is bandpass and real**

$$s_m(t) = \Re\left[A_m g(t) e^{j2\pi f_0 t}\right] \qquad [0, T]$$

$$= \Re\left[(A_m^{(I)} + jA_m^{(Q)}) g(t) e^{j2\pi f_0 t}\right]$$

Here,

$$k_1 \text{ bits mapped to } A_m^{(I)}$$

and

$$k_2 \text{ bits mapped to } A_m^{(Q)}$$

Thus,

$$M_1 = 2^{k_1}, \qquad M_2 = 2^{k_2}$$

and

$$M = M_1 M_2$$

Hence QAM can also be viewed as PAM-PSK constellation.

Total number of bits per symbol:

$$k = k_1 + k_2$$

$$k = \log_2(M_1 M_2) = \log_2 M$$

Expanding,

$$s_m(t) = A_m^{(I)} g(t) \cos(2\pi f_0 t) - A_m^{(Q)} g(t) \sin(2\pi f_0 t)$$

Proceeding similarly as for PSK,

Lowpass signal:

$$s_{ml}(t) = A_m g(t)$$

where

$$A_m = A_m^{(I)} + jA_m^{(Q)}$$

and

$$E_{sml} = \int_0^T |A_m|^2 g^2(t)\, dt = |A_m|^2 E_g$$

Lowpass basis:

$$\phi_l(t) = \frac{s_{ml}(t)}{\sqrt{E_{sml}}} = \frac{A_m g(t)}{|A_m|\sqrt{E_g}}$$

Since $g(t)$ is real,

$$\phi_l(t) = \frac{g(t)}{\sqrt{E_g}}$$

Only one lowpass basis exists.

Thus,

$$s_{ml}(t) = A_m\sqrt{E_g}\phi_l(t)$$

$$= (A_m^{(I)} + jA_m^{(Q)})\sqrt{E_g}\phi_l(t)$$

Bandpass basis functions:

$$\phi^I(t) = \sqrt{\frac{2}{E_g}}g(t)\cos(2\pi f_0 t)$$

$$\phi^Q(t) = -\sqrt{\frac{2}{E_g}}g(t)\sin(2\pi f_0 t)$$

Therefore,

$$s_m(t) = \frac{s_{ml}^{(I)}}{\sqrt{2}}\phi^I(t) + \frac{s_{ml}^{(Q)}}{\sqrt{2}}\phi^Q(t)$$

Since

$$s_{ml}^{(I)} = A_m^{(I)}\sqrt{E_g}$$

and

$$s_{ml}^{(Q)} = A_m^{(Q)}\sqrt{E_g}$$

we obtain

$$s_m(t) = A_m^{(I)}\sqrt{\frac{E_g}{2}}\,\phi^I(t) + A_m^{(Q)}\sqrt{\frac{E_g}{2}}\,\phi^Q(t)$$

Hence constellation coordinates are

$$\left\{A_m^{(I)}\sqrt{\frac{E_g}{2}},\, A_m^{(Q)}\sqrt{\frac{E_g}{2}}\right\}$$

## Symbol Energy

Signal vector:

$$\vec{s}_m = \left[ A_m^{(I)} \sqrt{\frac{E_g}{2}},\ A_m^{(Q)} \sqrt{\frac{E_g}{2}} \right]$$

Hence,

$$E_m = \|\vec{s}_m\|^2$$

$$= \frac{E_g}{2} \left[ (A_m^{(I)})^2 + (A_m^{(Q)})^2 \right]$$

## Minimum Euclidean Distance

Distance between two constellation points:

$$d_{mn} = \sqrt{\|\vec{s}_m - \vec{s}_n\|^2}$$

For adjacent points,

$$(A_m^{(I)} - A_n^{(I)}) = 2$$

or

$$(A_m^{(Q)} - A_n^{(Q)}) = 2$$

Thus,

$$d_{\min} = \sqrt{\frac{E_g}{2} \left[ (-1-1)^2 + (1-1)^2 \right]}$$

$$= \sqrt{2E_g}$$

## Average Energy for Rectangular QAM

For rectangular (square) constellation,

$$\sqrt{M} \times \sqrt{M} = M$$

with

$$M = 2^k$$

and amplitudes on each axis:

$$\pm 1, \pm 3, \ldots, \pm(\sqrt{M} - 1)$$

Average energy:

$$E_{\text{avg}} = \sum_{m=1}^{M} E_m P(s_m)$$

For equiprobable symbols,

$$P(s_m) = \frac{1}{M}$$

Thus,

$$E_{\text{avg}} = \frac{1}{M} \sum_{m=1}^{M} E_m$$

Substituting for $E_m$,

$$E_{\text{avg}} = \frac{E_g}{2M} \sum \left[ (A_m^{(I)})^2 + (A_m^{(Q)})^2 \right]$$

For square constellation,

$$E_{\text{avg}} = \frac{E_g}{2M} \left[ 2 \sum_{n=1}^{\sqrt{M}} A_n^2 \right]$$

Using PAM result,

$$\sum A_n^2 = \frac{\sqrt{M}(M-1)}{3}$$

Hence,

$$E_{\text{avg}} = \frac{(M-1)}{3} E_g$$

Average bit energy:

$$E_{b,\text{avg}} = \frac{E_{\text{avg}}}{\log_2 M}$$

Thus,

$$E_g = \frac{3 \log_2 M}{M-1} E_{b,\text{avg}}$$

Substituting into $d_{\text{min}}$:

$$d_{\text{min}} = \sqrt{2E_g}$$

$$= \sqrt{\frac{6 \log_2 M}{M-1} E_{b,\text{avg}}}$$

However for non-rectangular constellation, we need to work out after obtaining coordinates of symbol points in constellation.

The modulator consists of a vector mapper, which maps each of the $M$ messages onto a constellation of size $M$, followed by a two-dimensional vector-to-signal mapper.

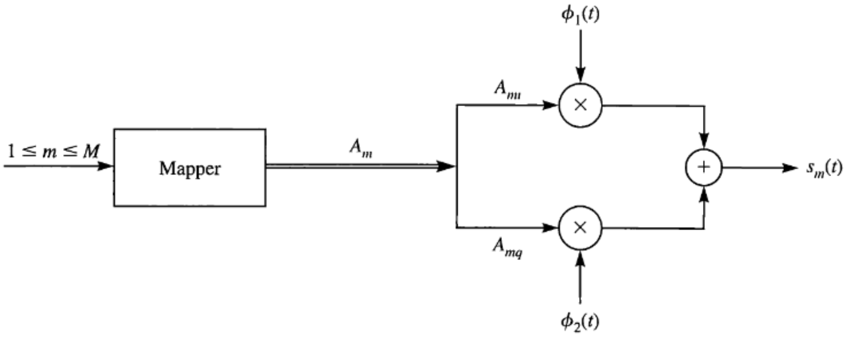



Figure 16: General QAM Modulator

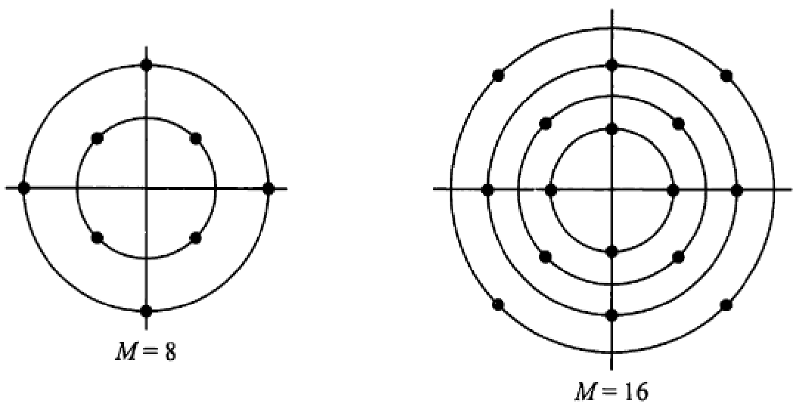



Figure 17: Example of combined PAM-PSK constellations

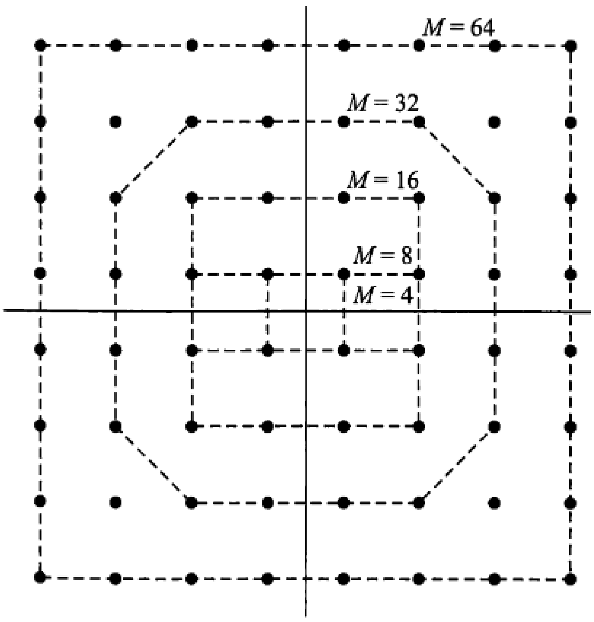



Figure 18: Rectangular QAM Constellations

| **Signaling** | $\mathbf{s_m(t)}$ | $\mathbf{s_m}$ | $\mathbf{E_{avg}}$ | $\mathbf{E_{b,avg}}$ | $\mathbf{d_{min}}$ |
|---|---|---|---|---|---|
| Baseband PAM | $A_m p(t)$ | $A_m\sqrt{E_p}$ | $\frac{2(M^2-1)}{3}E_p$ | $\frac{2(M^2-1)}{3\log_2 M}E_p$ | $\sqrt{\frac{6\log_2 M}{M^2-1}E_{\text{bavg}}}$ |
| Bandpass PAM | $A_m g(t)\cos(2\pi f_c t)$ | $A_m\sqrt{\frac{E_g}{2}}$ | $\frac{M^2-1}{3}E_g$ | $\frac{M^2-1}{3\log_2 M}E_g$ | $\sqrt{\frac{6\log_2 M}{M^2-1}E_{\text{bavg}}}$ |
| PSK | $g(t)\cos\left(2\pi f_c t+\frac{2\pi}{M}(m-1)\right)$ | $\sqrt{\frac{E_g}{2}}\begin{pmatrix}\cos\frac{2\pi}{M}(m-1)\\ \sin\frac{2\pi}{M}(m-1)\end{pmatrix}$ | $\frac{1}{2}E_g$ | $\frac{1}{2\log_2 M}E_g$ | $2\sqrt{\log_2 M\sin^2\left(\frac{\pi}{M}\right)E_{\text{bavg}}}$ |
| QAM | $A_{mi}g(t)\cos(2\pi f_c t)-A_{mq}g(t)\sin(2\pi f_c t)$ | $\sqrt{\frac{E_g}{2}}\begin{pmatrix}A_{mi}\\ A_{mq}\end{pmatrix}$ | $\frac{M-1}{3}E_g$ | $\frac{M-1}{3\log_2 M}E_g$ | $\sqrt{\frac{6\log_2 M}{M-1}E_{\text{bavg}}}$ |

Table 1: Comparison of PAM, PSK, and QAM

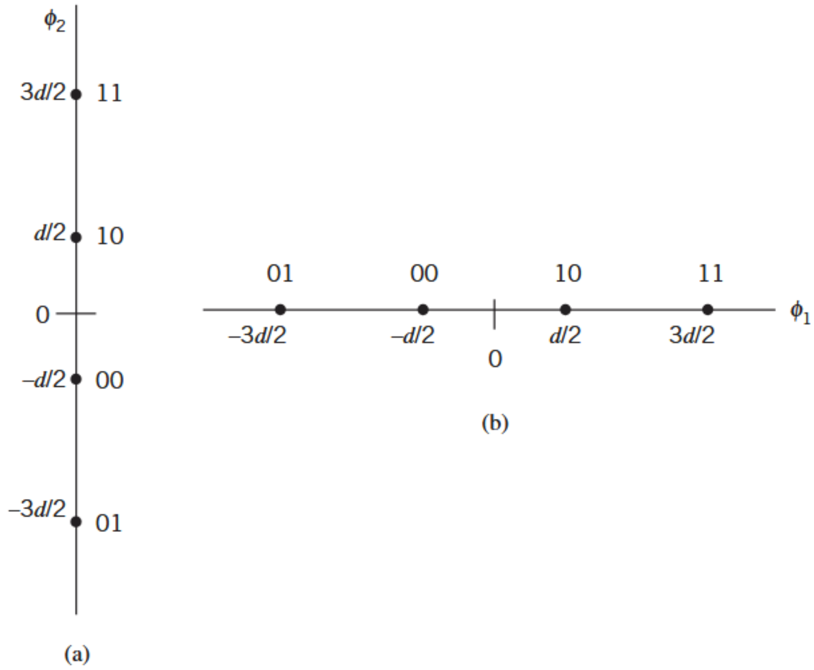



Figure 19: The two orthogonal constellations of the 4-ary PAM. (a) Vertically oriented constellation. (b) Horizontally oriented constellation. As mentioned in the text, we move top-down along the $\phi_2$-axis and from left to right along the $\phi_1$-axis.

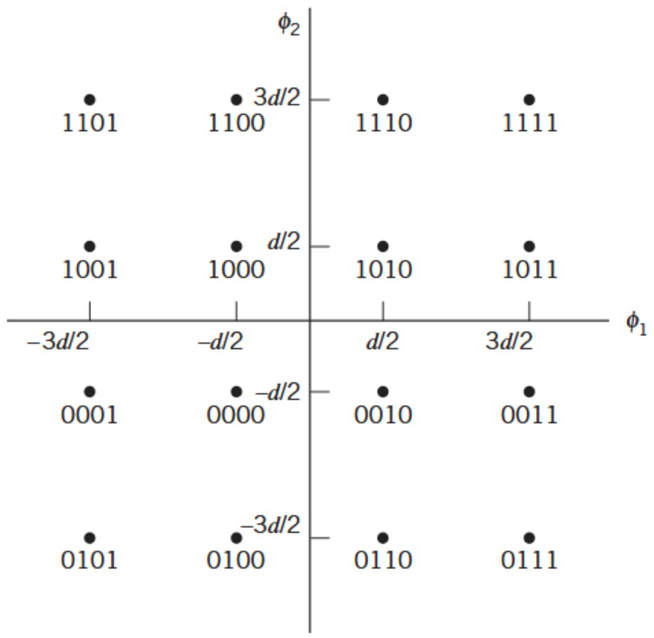



Figure 20: (a) Signal-space diagram of M-ary QAM for M = 16; the message points in each quadrant are identified with Gray-encoded quadbits

# Summary of Bandlimited Modulation Schemes (PAM, PSK, QAM)

From bandpass PAM, PSK, and QAM discussions, it is clear that all these schemes have a general form:

$$s_m(t) = \Re\left[A_m g(t) e^{j2\pi f_c t}\right], \qquad m = 1, 2, \ldots, M \tag{86}$$

where $A_m$ is determined by the signalling scheme.

**M-PAM** $\rightarrow$ $A_m$ is real, $A_m = \pm 1, \pm 3, \ldots, \pm(M-1)$
**M-PSK:**$\rightarrow$ $A_m$ is complex, $A_m = e^{j\frac{2\pi}{M}(m-1)}$
**M-QAM:** $\rightarrow$ $A_m$ is a general complex number,$A_m = A_{mi} + jA_{mq}$
In QAM, amplitude and phase both carry information. The dimensionality of PAM is 1, and for PSK and QAM it is 2, and it is independent of $M$.

# 5. Binary Frequency Shift Keying (BFSK)

BFSK is a non-linear modulation scheme, where the amplitude of the modulated signal does not vary linearly with the message signal. These non-linear modulation schemes are power efficient but bandwidth inefficient. Examples: FSK, MSK, GMSK, constant-envelope modulation.

BFSK Signal

$$s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t), \qquad 0 \le t \le T_b \tag{87}$$

where $i = 1, 2$, and $E_b$ is the transmitted energy per bit.

$$f_i = \frac{n_c + i}{T_b}, \quad i = 1, 2 \tag{88}$$

Symbol 1 $\rightarrow s_1(t)$, Symbol 0 $\rightarrow s_2(t)$.

The FSK discussed here is a continuous-phase signal; phase continuity is always maintained, including inter-bit switching times.

$$\phi_i(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_i t), \qquad 0 \le t \le T_b \tag{89}$$

the coefficients:

$$S_{ij} = \int_0^{T_b} s_i(t)\phi_j(t)\, dt \tag{90}$$

$$S_{ij} = \int_0^{T_b} \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t) \sqrt{\frac{2}{T_b}} \cos(2\pi f_j t)\, dt \tag{91}$$

$$S_{ij} = \begin{cases} \sqrt{E_b}, & i = j \\ 0, & i \neq j \end{cases} \tag{92}$$

Binary FSK has a two-dimensional signal space with two message points.
The two message points are defined by the vectors

$$\mathbf{s}_1 = \begin{bmatrix} \sqrt{E_b} \\ 0 \end{bmatrix}, \qquad \mathbf{s}_2 = \begin{bmatrix} 0 \\ \sqrt{E_b} \end{bmatrix} \tag{93}$$

$$d_{\min} = \sqrt{2E_b} \tag{94}$$

## PSD of BFSK

The BFSK signal can be expressed as

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi f_c t \pm \frac{\pi t}{T_b}\right), \quad 0 \le t \le T_b \tag{95}$$

$$s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos\left(\pm\frac{\pi t}{T_b}\right) \cos(2\pi f_c t) - \sqrt{\frac{2E_b}{T_b}} \sin\left(\pm\frac{\pi t}{T_b}\right) \sin(2\pi f_c t) \tag{96}$$

$$s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos\left(\frac{\pi t}{T_b}\right) \cos(2\pi f_c t) \pm \sqrt{\frac{2E_b}{T_b}} \sin\left(\frac{\pi t}{T_b}\right) \sin(2\pi f_c t) \tag{97}$$

In the above equation, the "+" sign corresponds to transmitting symbol 0 and the "−" sign corresponds to transmitting symbol 1.

Also, the inphase component is completely independent of the input binary wave. It equals $\sqrt{\frac{2E_b}{T_b}} \cos\left(\frac{\pi t}{T_b}\right)$ for all time $t$, the PSD consists of two delta functions and weighted by a factor $\frac{E_b}{2T_b}$ occuring at $f = \pm\frac{1}{2T_b}$

The quadrature component is directly related to the input binary sequence. During the signalling interval $0 \le t \le T_b$, it equals $-g(t)$ when symbol is 1 and $+g(t)$ when symbol is 0, with g(t) is a pulse shaping function.

$$g(t) = \sqrt{\frac{2E_b}{T_b}} \sin\left(\frac{\pi t}{T}\right), \qquad 0 \le t \le T$$

Power spectral density of $g(t)$ is

$$\psi_g(f) = \frac{8E_bT_b\cos^2(\pi fT_b)}{\pi^2(4T_b^2f^2-1)^2}$$

The total PSD is the sum of the PSDs of inphase and quadrature components as given by

$$S_{xx}(f) = \frac{E_b}{2T_b}\left[\delta\left(f-\frac{1}{2T_b}\right)+\delta\left(f+\frac{1}{2T_b}\right)\right] + \frac{8E_bT_b\cos^2(\pi fT_b)}{\pi^2(4T_b^2f^2-1)^2}$$

- The baseband PSD of a BFSK signal with continuous phase ultimately falls off as the inverse fourth power of frequency.
- A BFSK signal with continuous phase does not produce as much interference outside the signal band of interest as a corresponding FSK signal with discontinuous phase does.
- This property of CPBFSK will be utilized in MSK, which will be discussed further in detail.

In BFSK,

$$f_1 = f_c + \frac{\Delta f}{2}, \qquad f_2 = f_c - \frac{\Delta f}{2}$$

$\Delta f \rightarrow$ frequency separation
Modulation index

$$h = \frac{\Delta f}{R_b} = T_b\Delta f$$

$$\Delta f_{\min} = \frac{1}{T_b} \qquad \text{for non-coherent FSK detection}$$

$$\Delta f_{\min} = \frac{1}{2T_b} \qquad \text{for coherent FSK detection}$$

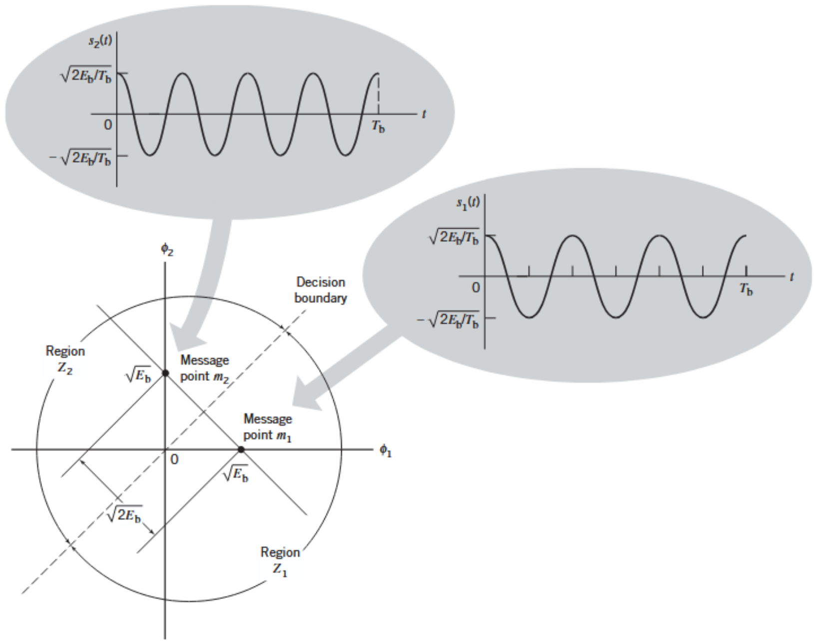



Figure 21: Signal-space diagram for binary FSK system. The diagram also includes example waveforms of the two modulated signals s1 (t) and s2 (t)

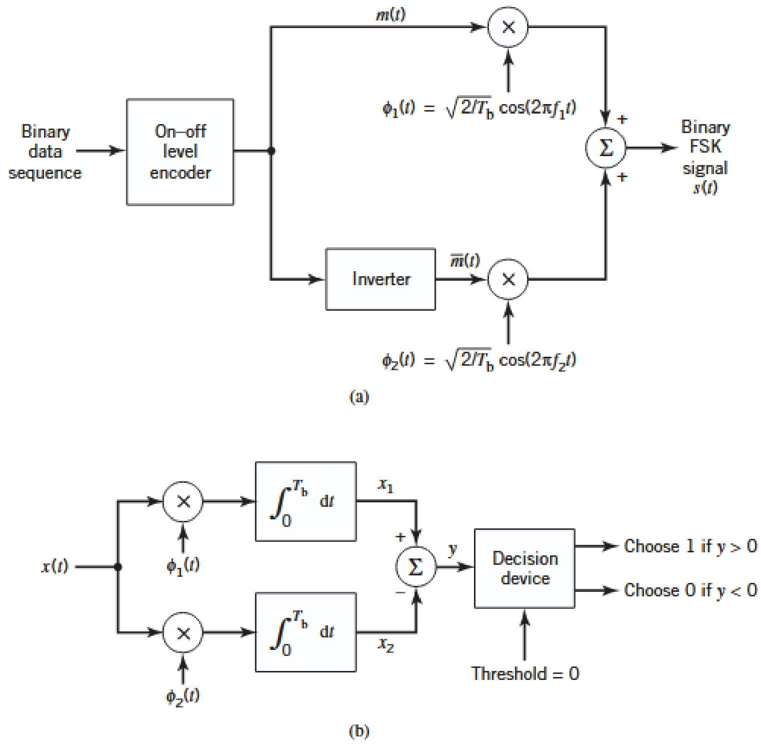



Figure 22: Block diagram for (a) binary FSK transmitter and (b) coherent binary FSK receiver

## Bandwidth Efficiency of $M$-ary FSK Signals

When the orthogonal signals of an $M$-ary FSK signal are detected coherently, the adjacent signals need only be separated from each other by a frequency difference $\frac{1}{2T_s}$ so as to maintain orthogonality. Hence, we may define the channel bandwidth required to transmit $M$-ary FSK signals as

$$B = \frac{M}{2T_s} \tag{98}$$

Since

$$T_s = kT_b = \frac{k}{R_b} \tag{99}$$

the bandwidth becomes

$$B = \frac{R_b M}{2\log_2 M} \tag{100}$$

The bandwidth efficiency of $M$-ary signals is therefore

$$\rho = \frac{R_b}{B} = \frac{2\log_2 M}{M} \tag{101}$$

We see that increasing the number of levels $M$ tends to increase the bandwidth efficiency of $M$-ary PSK signals, but it also tends to decrease the bandwidth efficiency of $M$-ary FSK signals. In other words, $M$-ary PSK signals are spectrally efficient, whereas $M$-ary FSK signals are spectrally inefficient.

### Bandwidth Efficiency Table

| $M$ | 2 | 4 | 8 | 16 | 32 | 64 |
|---|---|---|---|---|---|---|
| $\rho$ (bits/s/Hz) | 1 | 1 | 0.75 | 0.5 | 0.3125 | 0.1875 |

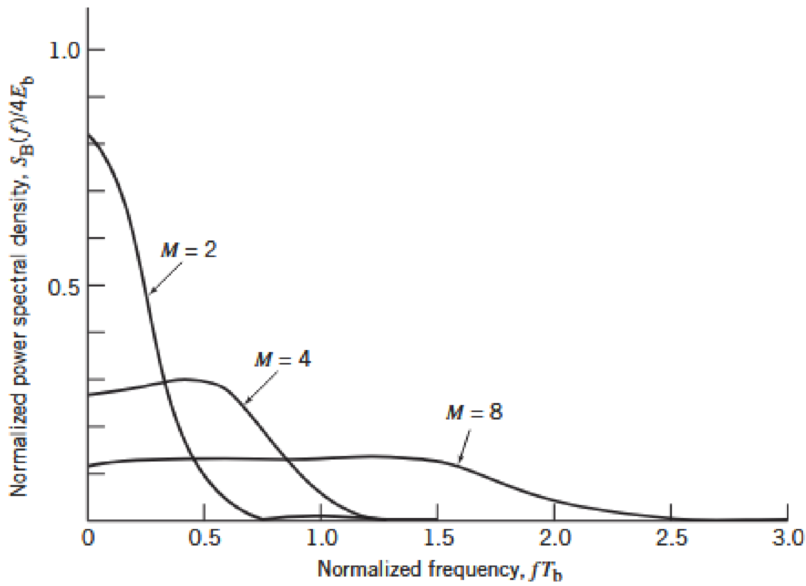



Figure 23: Power spectra of M-ary PSK signals for M = 2, 4, 8

# Noncoherent Detection

To tackle the signal detection problem given $x(t)$, we employ the generalized receiver, which consists of a pair of filters matched to the transmitted signals $s_1(t)$ and $s_2(t)$.

Because the carrier phase is unknown, the receiver relies on amplitude as the only possible discriminant. Accordingly, the matched-filter outputs are envelope-detected, sampled, and then compared with each other. If the upper path has an output amplitude $l_1$ greater than the output amplitude $l_2$ of the lower path, the receiver decides in favor of $s_1(t)$. If the converse is true, the receiver decides in favor of $s_2(t)$. When they are equal, the decision may be made by flipping a fair coin (i.e., randomly). In any event, a decision error occurs when the matched filter that rejects the signal component of the received signal $x(t)$ has a larger output amplitude (due to noise alone) than the matched filter that passes it.

# Binary Frequency-Shift Keying Using Noncoherent Detection

In binary FSK, the transmitted signal is

$$s_i(t) = \begin{cases} \sqrt{\dfrac{2E_b}{T_b}} \cos(2\pi f_i t), & 0 \leq t \leq T_b \\ 0, & \text{elsewhere} \end{cases} \tag{102}$$

where $T_b$ is the bit duration and the carrier frequency $f_i$ equals one of two possible values $f_1$ and $f_2$. To ensure that the signals representing these two frequencies are orthogonal, we choose

$$f_i = \frac{n_i}{T_b} \tag{103}$$

where $n_i$ is an integer. The transmission of frequency $f_1$ represents symbol 1, and the transmission of frequency $f_2$ represents symbol 0.

For the noncoherent detection of this frequency-modulated signal, the receiver consists of a pair of matched filters followed by envelope detectors. The filter in the upper path of the receiver is matched to $\cos(2\pi f_1 t)$, and the filter in the lower path is matched to $\cos(2\pi f_2 t)$ for the signaling interval $0 \leq t \leq T_b$. The resulting envelope detector outputs are sampled at $t = T_b$ and their values are compared. The receiver decides in favor of symbol 1 if $l_1 > l_2$ and in favor of symbol 0 if $l_1 < l_2$. If $l_1 = l_2$, the receiver simply guesses randomly.

The noncoherent binary FSK described here is a special case of noncoherent orthogonal modulation with $T = T_b$ and $E = E_b$, where $E_b$ is the signal energy per bit. Hence, the bit error rate (BER) for noncoherent binary FSK is

$$P_e = \frac{1}{2} \exp\left(-\frac{E_b}{2N_0}\right) \tag{104}$$

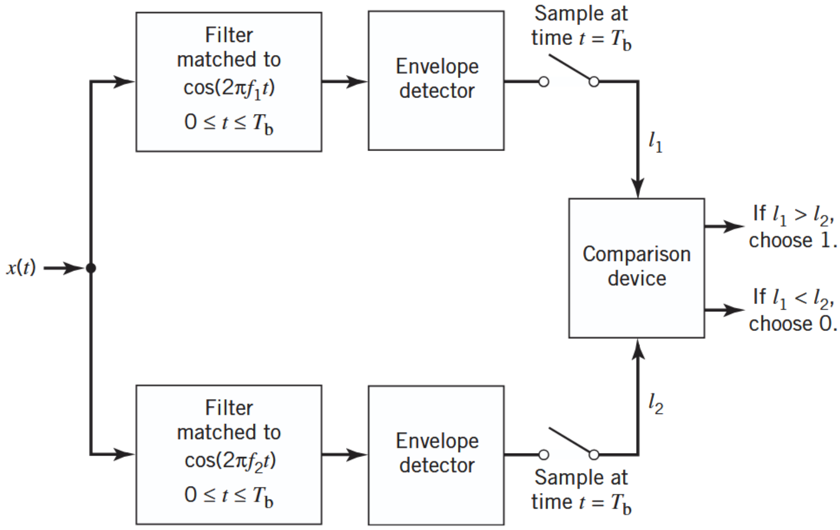



Figure 24: Noncoherent receiver for the detection of binary FSK signals

# Differential Phase-Shift Keying

We may view DPSK as the "noncoherent" version of binary PSK. The distinguishing feature of DPSK is that it eliminates the need for synchronizing the receiver to the transmitter by combining two basic operations at the transmitter:

- differential encoding of the input binary sequence, and
- PSK of the encoded sequence, from which the name of this new binary signaling scheme follows.

Differential encoding starts with an arbitrary first bit, serving as the *reference bit*; to this end, symbol 1 is used as the reference bit. Generation of the differentially encoded sequence then proceeds in accordance with a two-part encoding rule as follows:

1. If the new bit at the transmitter input is 1, leave the differentially encoded symbol unchanged with respect to the current bit.
2. If, on the other hand, the input bit is 0, change the differentially encoded symbol with respect to the current bit.

The differentially encoded sequence, denoted by $\{d_k\}$, is used to shift the sinusoidal carrier phase by zero and 180°, representing symbols 1 and 0, respectively. Thus, in terms of phase-shifts, the resulting DPSK signal follows the two-part rule:

1. To send symbol 1, the phase of the DPSK signal remains unchanged.
2. To send symbol 0, the phase of the DPSK signal is shifted by 180°.

# Illustration of DPSK

Consider the input binary sequence, denoted $\{b_k\}$, to be 10010011, which is used to derive the generation of a DPSK signal. The differentially encoded process starts with the reference bit 1. Let $\{d_k\}$ denote the differentially encoded sequence starting in this manner and $\{d_{k-1}\}$ denote its delayed version by one bit. The complement of the modulo-2 sum of $\{b_k\}$ and $\{d_{k-1}\}$ defines the desired $\{d_k\}$, as illustrated in Table 7.6. In the last line of this table, binary symbols 1 and 0 are represented by phase-shifts of 0 and $\pi$ radians.

Table 2: Illustrating the generation of DPSK signal

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $\{b_k\}$ | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| $\{d_{k-1}\}$ | | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| Differentially encoded sequence $\{d_k\}$ | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| Transmitted phase (radians) | 0 | 0 | $\pi$ | 0 | 0 | $\pi$ | 0 | 0 |

Basically, the DPSK is also an example of noncoherent orthogonal modulation when its behavior is considered over two-bit intervals; that is, $0 \le t \le 2T_b$. To elaborate, let the transmitted DPSK signal be

$$s_1(t) = \begin{cases} \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t), & symbol\ 1\ for \quad 0 \le t \le T_b \\ \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t), & symbol\ 0\ for \quad T_b \le t \le 2T_b \end{cases} \tag{105}$$

Suppose the input symbol for the second-bit interval $T_b \le t \le 2T_b$ is 0. According to part 2 of the DPSK encoding rule, the carrier phase is shifted by $\pi$ radians (i.e., 180°), thereby yielding the new DPSK signal

$$s_2(t) = \begin{cases} \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t), & symbol\ 1\ for \quad 0 \le t \le T_b \\ \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t + \pi), & symbol\ 1\ for \quad T_b \le t \le 2T_b \end{cases} \tag{106}$$

We now readily see that $s_1(t)$ and $s_2(t)$ are orthogonal over the two-bit interval $0 \le t \le 2T_b$, which confirms that DPSK is indeed a special form of noncoherent orthogonal modulation.

Hence, the bit error rate (BER) for DPSK is

$$P_e = \frac{1}{2}\exp\left(-\frac{E_b}{N_0}\right). \tag{107}$$

According to this formula, DPSK provides a gain of 3 dB over binary FSK using noncoherent detection for the same $E_b/N_0$.

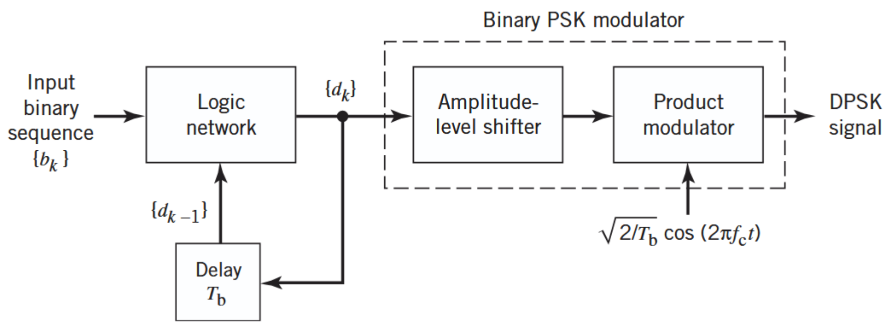



Figure 25: Block diagram of a DPSK transmitter.

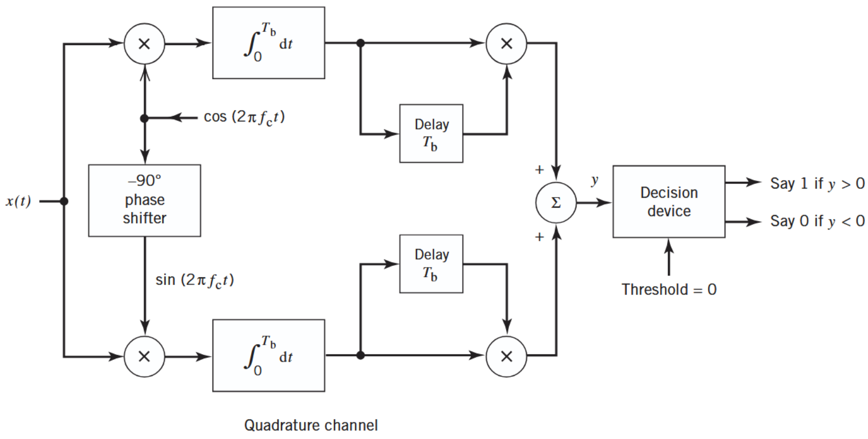



Figure 26: Block diagram of a DPSK receiver.